

Exploring equity standards for transport policy: A case study of ~~calculated~~ and objective accessibility, perceived accessibility, and employment outcomes in the Greater Toronto and Hamilton Area, Canada

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Abstract

The lack of standards for the equitable distribution of accessibility complicates efforts to address transport poverty. This study ~~identifies equity standards~~ presents a model-based approach for selecting equity standards. This is done by exploring the relationship between ~~calculated and perceived job accessibility for individuals living~~ objective accessibility and two employment-related variables for a sample of residents in the Greater Toronto and Hamilton Area, Canada. We use ~~calculated accessibility indicators~~ objective accessibility calculated for three modes of travel (active transportation, public transit, and ~~motorized modes~~ along private vehicles) and other individual-level variables to estimate ordinal models. ~~To control for endogeneity~~ Further, we use spatial filters as instrumental variables. ~~Two variables measure the perceptions of accessibility in the sample: to account for potential endogeneity~~ We examine the association between objective accessibility and the perception of having too few jobs available within an ideal commuting time and the perception of having to decline (a perceived accessibility indicator) and the reports of having declined jobs because of transportation constraints (an employment outcome indicator). The results indicate that higher ~~levels of active and transit accessibility tend to mitigate negative perceptions regarding these situations~~ transit accessibility is associated with more positive perceived accessibility, whereas associations with active accessibility depend on whether respondents travel actively. Based on the results of these models, we identify accessibility thresholds that reduce the likelihood of negative perceptions ~~for equity-deserving and outcomes for transport-disadvantaged groups~~. We ~~present three equity standards~~ propose three thresholds that we classify as conservative, intermediate, and progressive. ~~We believe that these standards can serve as a data-driven example of how~~ These thresholds are an example of a model-driven approach to support transportation planning in achieving more equitable outcomes, as well as for participatory processes to collectively define minimum or sufficient accessibility levels.

Keywords: benchmark, justice, ~~effect-job~~, egalitarian, sufficient

1. Introduction

Transportation equity refers to the distribution of the burdens and benefits of transportation systems among a population (Karner et al., 2020; Pereira et al., 2017). Burdens include health-related concerns arising from exposure to air pollution, noise, and traffic; sedentary lifestyles; and the risk of injury or death from collisions. In contrast, the primary benefit is the provision of accessibility (Pereira and Karner, 2021), understood as the ease of reaching (spatially distributed) opportunities and destinations (Páez et al., 2012;

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Hansen, 1959; Soukhov et al., 2025a), including employment, healthcare, educational services, and venues for social interaction.

Accessibility is one of the primary outcomes of spatial development (Páez et al., 2012). It is influenced by the spatial distribution of the population and opportunities and depends on the characteristics and performance of the transportation network. In other words, it is a product of both transportation systems and land use patterns (Wee and Geurs, 2011). There is a growing consensus among researchers and transportation planners that the goal of transport policy should be to improve accessibility (El-Geneidy and Levinson, 2022; Boisjoly and El-Geneidy, 2017). ~~This contrasts with traditional planning approaches that have prioritized mobility patterns that stimulate private automobile use and higher speeds as solutions to congestion and long commuting times. Mobility-centered policies often focus on metrics like increasing the number of trips, raising average speeds, and reducing congestion, treating mobility as an end in itself. However, this approach has proved problematic as movement becomes the measure of success, while leading to numerous other issues. In reality people travel to access opportunities, activities, or other people at their destinations, and so accessibility is seen as a superior indicator of performance (Pereira and Herszenhut, 2023).~~

~~Insufficient~~, since activity participation is one of the main ways in which transportation can help to promote a dignified life (Karner et al., 2020; Dos Santos et al.). In contrast, ~~insufficient~~ or inadequate accessibility can prevent individuals from ~~participating in location-based activities they wish to engage in for reasons engaging in activities that are meaningful to them, for reasons that are~~ beyond their control, ~~in a process~~. This process is described in the literature as Transport-Related Social Exclusion (TRSE) (Litman, 2024; Lucas, 2012; Preston and Rajé, 2007; Luz and Portugal, 2022; Paez et al., 2009). Insufficient accessibility, when combined with other forms of social and economic disadvantage, can result in transportation poverty (Lucas, 2019; Luz and Portugal, 2022): the failure of transportation and land use systems to provide access to essential services ~~or employment, employment, and other activities~~. Transportation poverty increases vulnerability among individuals who already face challenges associated with age (van Dülmen et al., 2022), income (Allen and Farber, 2019), race (Awaworyi Churchill, 2020), gender (Murphy et al., 2023), disability (Upham et al., 2022), among other factors (Kim et al., 2024). With the suburbanization of low-income populations, it is estimated that over one million Canadians live in transport poverty (Allen and Farber, 2019).

~~Complicating efforts to address~~ A challenge with the use of accessibility as a measure of transportation poverty is the lack of standards for the equitable distribution of accessibility (Soukhov et al., 2025b). Additionally, there is an absence of transportation data in the form of open data products to support urban planners ~~(Arribas-Bel et al., 2021; Behrendt and Sheller, 2021; Páez, 2021)~~ (Arribas-Bel et al., 2021; ?; Páez, 2021), and transportation planners receive limited formal guidance in assessing the equity implications of their projects (Miller, 2018; Palm et al., 2023; te Brömmelstroet et al., 2016). Furthermore, relatively few studies explicitly link accessibility levels with the experience of accessibility, such as employment opportunities declined or accepted due to transport conditions, or the ease to reach employment opportunities within ideal commuting times. Combined, this results in a gap in the development and implementation of accessibility standards to ensure equity in transportation planning and policy.

This study is guided by the following research questions: How ~~does calculated job accessibility do~~ objective job accessibility indicators, i.e., calculated based on spatial and transport data, relate to perceived accessibility ~~of employment opportunities and their and to self-reported employment~~ outcomes? And could a given level of accessibility serve as a standard or goal for equitable transport planning? ~~To answer these questions, this study has the general objective of proposing equity standards for transportation planning based on the relationship between calculated job accessibility and~~ This study proposes a model-based methodology for defining equity standards in transportation planning. We obtain these standards from the estimated relationships between objective job accessibility, people's ~~pereceptions of employment reachability~~ perceived accessibility, and a self-reported employment outcome.

~~Our research advances the literature on the equitable distribution of transportation benefits by demonstrating~~ While prior studies have examined the association between objective and perceived accessibility (Pot et al., 2021; Lättman et al., 2021), and between objective accessibility and employment outcomes (Bastiaanssen et al., 2021; Duarte et al., 2023; Jin and Paul, 2023), few have used these relationships to derive accessibility thresholds (e.g., Negm and El-Geneidy (2026)). Grounded in a transport equity perspective, the novelty of our work is the use of a model-based approach

to identify calculated accessibility standards based on their relationship with perceived accessibility—more specifically, with perceptions about the ease to reach employment opportunities. This research is made possible by the release of Canada’s first National Survey of Transport Poverty (NSTP) (Tiznado-Aitken et al., 2025). The proposed standards are derived define accessibility thresholds conditioned on individual characteristics, allowing standards to be defined from the standpoint of a chosen reference profile. Specifically, we define accessibility thresholds as the minimum levels at which the reference profile becomes more likely to give a non-negative response than a negative one, or, under a more demanding definition, a positive response than a negative one. Although these thresholds are obtained at the neighbourhood level, while perceptions reflect the experiences of respondents to the NTPS that reside in those areas. In this way, these thresholds can assist policymakers in interpreting how different levels of calculated accessibility are associated with residents’ perceived ability to reach employment opportunities under existing social and spatial constraints. This increases the potential for government adoption and enhances the comprehensibility and relevance of such standards for community members and those experiencing unequal accessibility. Furthermore, identifying sufficient job accessibility levels can help assess and highlight areas at higher risk of transport poverty, particularly those with a higher concentration of equity-deserving populations they indirectly incorporate individual-level characteristics, since they are derived from individuals’ self-reported perceptions and outcomes.

This study is guided by an understanding of accessibility as a means through which transportation can foster human dignity (Pereira and Karner, 2021; Dos Santos et al.), combined with an egalitarian and sufficientarian view of transportation equity. From this perspective, a fair transportation system should guarantee a minimum, sufficient level of accessibility to all members of society, with this sufficient level defined from the standpoint of those historically and currently facing the greatest transportation disadvantage (Martens and Golub, 2021).

In addition, to facilitate collaboration and further analysis and to ensure the transparency and replicability of our study, we developed this paper using literate programming, with the data analysis code accessible through our GitHub page (<https://github.com/dias-bruno/NSTP/>), in alignment with best practices in spatial data science (Páez, 2021; Arribas-Bel et al., 2021). After this introduction, we structured this article as follows: Section 2 provides a brief theoretical background on objective and perceived accessibility, on the identified connections between objective job accessibility and employment outcomes, and on the identification of equity standards based on accessibility. Section 3 presents the materials and methods, covering the study area and data sources, the method used to calculate accessibility, how we modelled the relationship between objective accessibility and the dependent variables, the process used to identify equity standards, and how we measured transportation equity based on these standards. Section 4 presents the results. Section 5 critically analyzes the paper and presents its limitations. Finally, Section 6 concludes with final considerations.

2. Theoretical ~~Background~~background

2.1. ~~Calculated~~Objective and perceived accessibility

The most commonly used accessibility measures are based on the gravity model of Hansen (1959)and make use of spatial data. This model considers the number of opportunities that a person can reach to be proportional to the sum of opportunities in each destination, discounted by the travel cost to reach that destination. This type of metric is classified as a ~~location-based~~place-based measure (Pereira and Herszenhut, 2023), representing a characteristic of a given location that assumes that all people in this location have equal access to the spatially distributed opportunities. However, they do. For relying on the use of spatial and transport data to create indicators, these measures are referred to as calculated, spatial, location-based, or objective accessibility; hereafter, we use the term *objective accessibility*.

However, objective accessibility does not take into account the individual characteristics of people.

The perceived accessibility, on the other hand, refers to the perception of people of how easily they can reach opportunities (Lättman et al., 2016; Pot, 2025; Mehdizadeh et al., 2025), resulting from not only the interaction of land-use and transport infrastructure (Geurs and van Wee, 2004), but also from travel options (e.g., having a car or a transit pass), travel experiences, and individual characteristics and

preferences. Often, perceived accessibility deviates from ~~place-based-calculated-objective~~ accessibility, with some research indicating that ~~calculated-objective~~ accessibility might not be able to capture the multiple ways in which people experience a lack of access to opportunities (Zhu and Lucas, 2025). Explanations for these differences are partially attributed to differences in individuals’ travel attitudes, capabilities, and preferences (De Vos et al., 2025; van der Vlugt et al., 2022). In addition, objective measures often rely on travel time as the only impedance factor, while other factors that might also shape individuals’ perceptions of the ease or difficulty of reaching destinations are not considered, such as safety, comfort, and even enjoyment of the trip (Handy and Clifton, 2001; Handy, 2020).

~~Even though, as demonstrated in the following subsection, calculated-~~Another factor that might impact the relationship between objective and perceived accessibility is adaptive preference, in which political oppression, poverty, and adverse social conditions distort the preferences of individuals living in deprived circumstances (Baber, 2007). In this situation, someone with restricted financial resources living in an area with low (objective) accessibility might adapt their expectations and preferences to align with their circumstances, and might therefore not consider their accessibility levels to be insufficient (Negm et al., 2025).

Residential self-selection can also impact the relationship between perceived and objective accessibility, in which people may choose their residential locations based on travel needs, abilities, and travel-related attitudes (Cao et al., 2009). In fact, residential location decisions differ across income groups, impacting the relationship between objective job accessibility and employment status (Hu, 2017). For instance, high-income individuals often prioritize living in places with low job accessibility because they prefer suburban housing amenities, such as green spaces and larger homes, over proximity to work, accepting longer commutes because they can afford - and might even prefer to have - a car. As a consequence, their perceived ability to reach employment opportunities may be defined less by their neighbourhood’s job accessibility and more by their resources and preferences. On the other hand, low-income individuals might prefer to live close to employment hubs to maximize their chances of having a job, while reducing commuting and housing costs.

Objective accessibility is still commonly used to evaluate the effect that the interaction between land-use characteristics and the transportation system has on perceived job accessibility and on employment outcomes. In addition, existing studies on perceived accessibility seem to have difficulties in creating standardized definitions and measurement tools, contributing to the somewhat limited knowledge on the topic (Negm et al., 2025). In this context, connecting ~~calculated-objective~~ to perceived accessibility helps to capture non-spatial and employment-related barriers that are not visible in place-based measures alone, since the ~~calculated-objective~~ accessibility might be perceived differently even between people living within that same neighbourhood. In this paper, ~~as suggested by Pot et al. (2025),~~ we define sufficient levels of accessibility based on the relationship between ~~calculated-objective~~ accessibility and people’s perception of ~~employment-reachability~~ job accessibility and employment outcomes, while also controlling for individual characteristics.

2.2. Job accessibility and employment outcomes

Studies connecting employment outcomes to job accessibility date back to the 1960s, when John Kain (1968) proposed the spatial mismatch hypothesis while analyzing Detroit and Chicago, USA. The author hypothesized that the low employment rates among Black Americans in inner cities resulted from employment decentralization and the difficulty faced by the Black population in moving out of these areas, primarily due to racial discrimination and segregation.

Since Kain (1968), studies linking ~~calculated-objective~~ job accessibility and different employment outcomes have presented mixed results. Bania et al. (2008) found no significant effect for job access when examining its relationship with employment, earnings, hourly wage, and weekly hours worked for women leaving welfare in Ohio, USA. Delmelle et al. (2021) also found that changes in car accessibility had no significant impact on employment rates for different population income groups in the Metropolitan Area of Charlotte, North Carolina, USA, arguing that the spatial separation between lower-wage workers and their workplaces might already be relatively small. Studying Los Angeles, USA, Merlin and Hu (2017) found mixed-effects of job accessibility on employment status, with a stronger association for more highly educated groups. The authors argued that lower-educated groups may be more limited by non-spatial factors

that impede their employment opportunities, such as qualifications~~and social skills~~, labour market readiness, and a lack of social networks. On the other hand, for workers with a higher level of education, the time and monetary costs of spatial separation itself may play a relatively more important role, so accessibility measures may have a greater impact on their decisions and ability to enter the labour market.

In contrast, other studies found significant effects of job accessibility on employment outcomes (Bastiaanssen et al., 2021; Duarte et al., 2023; Jin and Paulsen, 2018; Johnson et al., 2017; Matas et al., 2010). Jin and Paulsen (2018) used a difference-in-differences model for the Chicago Metropolitan Area, finding that increasing job accessibility between 2010 and 2017 was associated with reduced unemployment rates. Duarte et al. (2023) found that job accessibility had a positive effect on participation in the labour market for women, a negative effect on informal employment for women, and a positive effect on the probability of being employed for both women and men. More recently, Bastiaanssen et al. (2025) examined differential employment effects of job accessibility by public transport in combination with the bicycle for different educational groups at the national level in the Netherlands. They found that jobs for lower-educated groups are often located outside areas of high accessibility, thereby reducing job accessibility among these groups, while jobs for higher-educated groups tend to be concentrated in and around downtowns and are associated with greater accessibility. In this study, the assumed connection between the spatial dimension of accessibility and ~~the perceived~~ people's perceived accessibility and employment outcomes is the basis for the identification of equity standards.

2.3. Equity ~~Standards~~standards

An equity standard, when operationalized, defines the level of fairness to be adopted and the acceptable inequality (Soukhov et al., 2025b). Accessibility standards typically combine a concept of fairness with threshold criteria, reflecting both vertical and horizontal equity approaches. Vertical equity is often operationalized through opportunity-based standards defined for specific population groups (e.g., minimum access to opportunities within a 10-minute walking trip for older adults), whereas horizontal equity commonly appears in the use of uniform travel impedance thresholds applied across all potential users (e.g., a maximum acceptable distance to a transit stop).

~~Yenisetty & Bahadure (2020)~~ Yenisetty and Bahadure (2020) set an opportunity-based standard, considering a distance of more than 1,200 meters between amenities, services, and facilities by public transport to be unfair when analyzing cities in India. ~~Tiznado-Aitken (2018)~~ Tiznado-Aitken et al. (2018) defined unfair cases as those where bus stops and metro stations are located more than 1 kilometer away. ~~Sun & Thakuriah (2021)~~ Sun and Thakuriah (2021) adopted 400 meters as the maximum acceptable walking distance to bus and tram stops, and 800 meters to ferry stations in England. In another study, ~~Peunghumsai et al. (2020)~~ Peunghumsai et al. (2020) identified opportunity standards through the ratio of public transport supply to demand, considering conditions fair when demand is greater than supply.

The use of different thresholds reflects the idiosyncratic manner in which standards are developed, and the rationale for choosing a certain distance to, for example, bus stops or train stations, is often unclear. In addition, the use of distance-based thresholds may be problematic, since the time required to cross a certain distance can vary considerably within a city, even for the same mode of transportation. To address this problem, we suggest standards based on key thresholds of ~~calculated~~ objective accessibility.

However, beyond making explicit how standards are defined, it is also important to make transparent the moral principles that guide their definition, because different theories of justice lead to different perspectives on how policies should manage the distribution of transport-related benefits. Pereira and Karner (2021) argue that transportation equity research benefits from linking two justice theories: Rawlsian egalitarianism (Rawls, 1999) and the Capability Approach (Sen, 1979, 1999, 2005, 2009). Rawls' Difference Principle holds that inequalities are only justifiable if they benefit the least advantaged members of society, centred on the fair distribution of primary social goods - resources and opportunities that every person needs to pursue their life plans. Sen builds on this, arguing that the focus should shift from primary goods to human capabilities: the opportunities available to individuals and their freedom to choose and act, recognizing that the same resource can translate into different real opportunities depending on individual circumstances. Within transportation research, accessibility to opportunities can therefore be understood as a capability that

enables participation in social, economic, and community life (Pereira et al., 2017; Luz and Portugal, 2022)

Both theories inform two moral perspectives commonly applied, though rarely made explicit, in transportation equity research (Martens, 2016a; Wee and Geurs, 2011). The sufficientarian perspective asks why some groups or areas lack sufficient access, building on the idea of a minimum threshold of accessibility required to meet basic needs, with priority given to those falling below it. The egalitarian perspective instead focuses on relative differences in accessibility between groups or areas, treating unequal access as unfair regardless of whether a group’s absolute access level meets a minimum threshold (Pereira and Karner, 2021). Applying either perspective therefore requires identifying which groups are of concern: the sufficientarian view asks who falls below the threshold, while the egalitarian view asks between whom access differs.

In practice, this attention has centred on disadvantaged population groups, including those affected by low income, limited ability, and historical discrimination, which have become the focus of transportation research (Lucas, 2019). This is due to the potential of historical discriminatory policies and other disadvantages to negatively impact their transport experiences (Pereira and Karner, 2021). Additionally, these groups have often been excluded from the transportation planning process, resulting in their needs frequently being not recognized. Within studies that investigated associations between objective accessibility and employment outcomes, relevant groups include youth, adults, women, children, low-income individuals, zero-vehicle households, people with disabilities, single-parent households, refugees, and racialized populations (Bania et al., 2008; Hu, 2022).

In this study, we recognize accessibility as a human capability (Sen, 2005, 1999; Pereira et al., 2017; Luz and Portugal, 2022), something required for a dignified life (Karner et al., 2020; Dos Santos et al.), and one of the main benefits of transportation to be distributed across society in order to increase people’s freedom. Following the sufficientarian perspective, we define equity standards as minimum accessibility thresholds required to reduce the likelihood of negative perceptions and outcomes. Following an egalitarian view, these thresholds are defined with particular attention to transport-disadvantaged population groups. In doing so, the paper presents an analysis that identifies minimum acceptable levels of accessibility, discusses inequities in these values, and presents a generalizable methodology for other geographical areas, populations and opportunity types.

3. Materials and ~~Methods~~ methods

Our approach for obtaining equity standards of accessibility based on the relationship between ~~calculated measures and objective measures~~, people’s perceptions, and employment outcome can be synthesized in the following main steps: modelling job accessibility, predicting individual perception ~~using calculated and outcome using objective~~ accessibility, setting equity standards for transportation, measuring transportation equity by using Foster-Greer-Thorbecke (FGT) measures (Foster et al., 2010), ~~and defining priority geographical areas for policy interventions based on their Accessibility Fairness Index (AFI) (Martens, 2016b)~~.

3.1. Study ~~Area~~ area and ~~Data Sources~~ data sources

The study area for this research is the metropolitan region of the Greater Toronto Area and the Hamilton Area (GTHA), in Ontario, Canada. More specifically, the study area refers to the Census Divisions of the regional municipalities of Hamilton, Halton, Peel, Toronto, York, and the Durham subdivisions within the Toronto Census Metropolitan Area (Figure 1). According to the latest census (Canada, 2023), over 7.2 million residents live in this area, which is spread across around 7,300 square ~~kilometres~~ kilometres.

Regarding the primary data sources, this study uses data from two different sources: the latest Canadian Census (2021) from Statistics Canada to calculate objective accessibility; and the National Survey of Transport Poverty (NSTP) from the Mobilizing Justice Partnership (Tiznado-Aitken et al., 2025) to obtain variables related to individuals’ perceived accessibility and employment outcomes. The Census was selected due to its various socio-economic and demographic variables and due to the “Commuting to Work” questionnaire, allowing the identification of individuals in the labour force and including variables required to

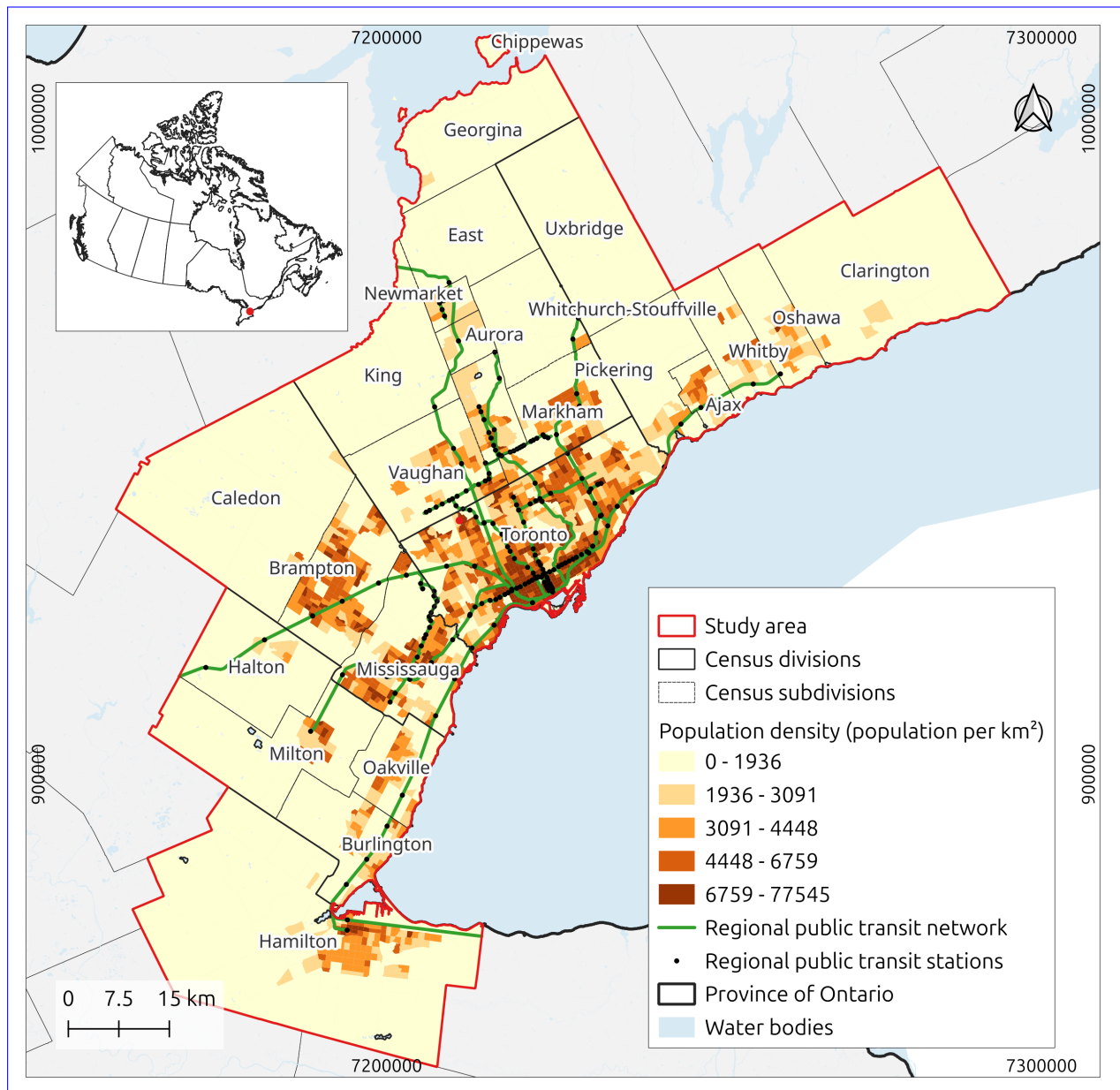


Figure 1: Map of the study area ([population density is classified using quintile breaks](#)).

Weighted summary statistics of the outcome variables.

Table 1: Weighted summary statistics of the dependent variables.

To what extent do you agree with the following statements?	N = 3,576,893
There are too few jobs available for me in my ideal commuting time	58.84 (27.71)
I have had to decline employment opportunities because of my transport situation	41.00 (31.62)

¹ Mean (SD)

calculate job accessibility, such as place of work, commuting modes, and commuting duration. The Census also provides standardized spatial datasets with statistical and administrative geographical boundaries, including the Census Tract (CTs), which are used in this study as the spatial units used for calculating accessibility measures (obtained using the `{cancensus} cancensusR R-package package` (von Bergmann et al., 2021)). According to the Census (Canada, 2023), 3,868,120 individuals in the labour force reside in the study area, competing for a total of 3,043,730 job opportunities.

The Census also provides the number of people in low income, which we use to assess accessibility gaps among low-income residents. Low income is defined using the Low-Income Measure, After-Tax (LIM-AT), a relative threshold set at 50% of the median adjusted after-tax household income, where each household’s after-tax income is adjusted by dividing it by the square root of household size. According to the 2021 Census, around 10% of the study area population lives in low income.

The second source of data, NSTP (Tiznado-Aitken et al., 2025), is a large-scale survey designed to understand TRSE and transport poverty in Canada. It is a representative sample of more than 27,000 people aged 15 years and older across the country, and it provides an unprecedented perspective look at the transportation challenges that Canadians face, especially among equity-deserving transport-disadvantaged groups whose experiences are often overlooked. The survey is openly available to researchers, decision-makers, and the public, and offers information on household dynamics, respondents’ perceptions, and individual-level socioeconomic data. The NSTP provides respondent weights to enable the creation of statistics that are consistent and representative of the Canadian population. From this survey we select variables corresponding to respondents’ perceptions of accessibility and employment outcomes, in addition to socioeconomic and demographic variables at the individual level. The responses are geocoded by place of residence and can be assigned to Census geographies to link them to calculated objective accessibility.

Not all questions from the NSTP were answered by all survey respondents. The questions were defined according to demographic and socioeconomic profiles and to previous responses. For this study, the NSTP only a sample of the NSTP was used. This sample corresponds to labour force respondents residing in the GTHA who also answered the questions displayed in Table 1, totalling 2,270 respondents that represent 3,576,893 individuals in the population (weighted sample) reported being able to work, whether they are employed or unemployed, and who are not retired, sick, or on parental leave. Using a continuum scale from 0 (completely disagree) to 100 (completely agree), the respondents answered to what extent they agreed with the statements in Table 1, namely “There are too few jobs available for me in my ideal commuting time” and “I have had to decline employment opportunities because of my transport situation”. In total, the NSTP sample refers to 2,270 respondents that represent 3,576,893 individuals in the labour force (weighted sample).

3.2. Calculating accessibility

In this study, we implement the multimodal spatial availability measure of Soukhov et al. (2024). This measure is a constrained indicator of accessibility that can account for multiple modes (see Soukhov et al., 2023, 2025a). This measure considers competition for opportunities, which prevents a single job opportunity to be available to more than one person. Competitive measures matter when considering employment outcomes because jobs are exclusive, and their use by one person prevents their use by another. In addition, this measure relies on a proportional allocation that balances the cost of travel and the population size, imposing a constrained calculation that allocates the opportunities to each origin. This ensures that, when all accessible

jobs are summed, the sum is equal to the total number of opportunities in the region. Lastly, this measure is a more interpretable indicator since it returns scores in the units of opportunities.

We calculated the multimodal spatial availability V_i^m using Equation (1):

$$V_i^m = \sum_{j=1}^J O_j F_{ij}^{tm}$$

where:

3.1.1. Objective accessibility inputs: labour force, job opportunities, and travel costs

V_i^m is the spatial availability of opportunities at origin i and travel mode m ; O_j is Calculating objective accessibility, as operationalized in this paper (see following section), requires three inputs: the number of opportunities at destination j ; F_{ij}^{tm} is a balancing factor allocating opportunities at j to origin i considering mode m each destination, the population competing for them at each origin, and the cost of travelling between origins and destinations, which is translated into travel impedance through a mode-specific function.

F_{ij}^{tm} incorporates both population demand and travel impedance, defined according to Equation 2:

$$F_{ij}^{tm} = \frac{F_i^{pm} \cdot F_{ij}^{cm}}{\sum_{m=1}^M \sum_{i=1}^N F_i^{pm} \cdot F_{ij}^{cm}}$$

The population-based allocation factor is obtained by:

$$F_i^{pm} = \frac{P_i^m}{\sum_m \sum_i P_i^m}$$

Where P_i^m represents the population at origin i using mode m . On its turn, the travel impedance allocation factor is given by:

$$F_{ij}^{cm} = \frac{f^m(c_{ij}^m)}{\sum_m \sum_i f^m(c_{ij}^m)}$$

Where:

- c_{ij}^m is the cost to travel between i and j for mode m ;
- $f^m c_{ij}^m$ is an impedance function representing the effect of travel cost.

Using microdata from the Census, we calculated for each CT the number of people in the labour force that commute to work by walking, cycling, ~~motorized modes~~ private vehicles (those who commute by car, truck, or van as a driver or passenger, and motorcycles or scooters), and public transit (those who commute by bus, subway, light rail, streetcar, commuter train, and passenger ferry). The labour force is defined as in the Canadian Census: civilians aged 15 and older who are employed or unemployed. We distributed remote workers according to the share of each transportation mode in each CT. Then, we summed the number of jobs in each CT considering the place of work mentioned by each worker.

We calculated the travel times between origins and destinations for each travel mode using the {r5r} R package (Pereira et al., 2021) with a street network obtained from OpenStreetMap. We used the General Transit Feed Specification (GTFS) obtained from the Canadian Public Transit Network Database (Canada, 2025) to retrieve the public transit schedule and coverage. Considering that our study area contains different transit agencies, we evaluated the GTFS files to identify a common date/week with public transit service. After this analysis, we set 8 a.m. (morning peak time) of December 22, 2025, as the trip departure time. Although this date falls during Christmas week, there were no differences in the public transit service schedules of the transit agencies due to the holiday, which makes this date representative of a regular weekday in terms of public transit travel time estimation. We also used Digital Elevation Models (DEM) from the Canadian Digital Surface Model (CDSM), with a spatial resolution of about 20 meters, to estimate

the effect of declivity on walking and cycling times. For cycling trips, we defined a standard level of traffic stress (LTS) as tolerable for most mainstream adults. LTS defines the level of comfort or danger that a street presents for cyclists based on speed limits, lane counts, and infrastructure. The standard LTS defined in {r5r} (Pereira et al., 2021) includes bike lanes with physical separation from multi-lane or higher-speed traffic. We set a maximum duration of 180 minutes (3 hours) for ~~motorized private vehicles~~ and public transit trips, with a maximum walking time of 5 minutes to reach the vehicle and 15 minutes to reach the public transit stop/station; for walking and cycling trips, we set a maximum duration of 90 minutes (one and a half hours). After estimating the travel times between CTs, we assumed that all intra-CT trip durations are equal to 0.1 minute.

Commute duration was used to calibrate Probability Density Functions (PDFs) to represent travel impedance, following the approach of Soukhov and Páez (2024), Verma and Ukkusuri (2025), and Dos Santos et al. (2026), where the impedance value represents the probability that the trip will occur given a ~~specific~~ specific travel time. We used the ~~{fitdistrplu}-s-fitdistrplus~~ R package (~~Delignette-Muller and Dutang, 2015a~~) (Delignette-Muller and Dutang, 2015b) and tested different functional forms (uniform, negative exponential, gamma, normal, and lognormal distributions) to select the best PDF for each mode of transportation and municipality, using ~~a-as the~~ selection criteria the lowest Akaike Information Criterion (AIC) value (Akaike, 1974). The estimated impedance functions appear in ~~Table-Appendix~~ A.1.

3.2. ~~Modeling the causal relationship between calculated and perceived~~ Calculating objective accessibility

In this study, we implement the multimodal spatial availability measure of Soukhov et al. (2024). This measure is a constrained indicator of accessibility that can account for multiple modes (see Soukhov et al., 2023, 2025a). This measure considers competition for opportunities, which prevents a single job opportunity from being available to more than one person. Competitive measures matter when considering employment outcomes because jobs are exclusive, and their use by one person prevents their use by another. In addition, this measure relies on a proportional allocation that balances the cost of travel and the population size, imposing a constrained calculation that allocates the opportunities to each origin. This ensures that, when all accessible jobs are summed, the sum is equal to the total number of opportunities in the region. Lastly, this measure is a more interpretable indicator since it returns scores in the units of opportunities.

We calculated the multimodal spatial availability V_i^m using Equation (1):

$$V_i^m = \sum_{j=1}^J O_j F_{ij}^{tm} \quad (1)$$

where:

- V_i^m is the spatial availability of opportunities at origin i and travel mode m ;
- O_j is the number of opportunities at destination j ;
- F_{ij}^{tm} is a balancing factor allocating opportunities at j to origin i considering mode m .

F_{ij}^{tm} incorporates both population demand and travel impedance, defined according to Equation 2:

$$F_{ij}^{tm} = \frac{F_i^{pm} \cdot F_{ij}^{cm}}{\sum_{m=1}^M \sum_{i=1}^N F_i^{pm} \cdot F_{ij}^{cm}} \quad (2)$$

The population-based allocation factor is obtained by:

$$F_i^{pm} = \frac{P_i^m}{\sum_m \sum_i P_i^m}$$

Where P_i^m represents the labour force at origin i using mode m competing for the opportunities O at the destination j .

In turn, the travel impedance allocation factor is given by:

$$\bullet \quad F_{ij}^{cm} = \frac{f^m(c_{ij}^m)}{\sum_m \sum_i f^m(c_{ij}^m)}$$

Where:

- c_{ij}^m is the cost to travel between i and j for mode m ;
- $f^m(c_{ij}^m)$ is an impedance function representing the effect of travel cost.

3.3. Modeling the relationship between objective accessibility, perceived accessibility, and employment outcomes

In this study, we aim to define minimum levels of ~~calculated~~ objective accessibility by identifying key thresholds in its relationship with perceived accessibility and employment outcomes. The underlying ~~assumption~~ hypothesis of this strategy is that ~~there is a connection between these two components of changes in the land-use and transport characteristics of a neighbourhood, captured and represented by objective accessibility, lead to changes in how residents perceive their ability to reach job opportunities and in their reported employment outcomes. We hypothesize that residents of neighbourhoods with more (objective) accessibility, such that changes in calculated accessibility would result on changes in the distribution of perceived accessibility. This implies a causal relationship between calculated (the cause or the treatment) and perceived accessibility (the outcome), rather than merely their statistical correlation. Therefore, we hypothesise that higher values of calculated accessibility will increase the probability of being in more positive perception categories, and that perceptions of accessibility can be connected to multimodal spatial availability indicators are less likely to perceive job scarcity or to report having declined job offers due to transport constraints, independent of their individual demographic and socioeconomic profiles. In our approach, we use objective accessibility not as a causal exposure in itself, but as a proxy for the underlying land-use and transport conditions that, as we hypothesize, shape individual perceptions and outcomes.~~

Following the steps suggested by Huntington-Klein (2021), we draw a ~~causal diagram~~ Directed Acyclic Graph (DAG) to graphically represent ~~the assumed causal relationship between calculated and perceived accessibility~~ our hypothesis (Figure 2). ~~Causal diagrams~~ DAGs show a chain of hypothesized causal effects by using nodes to symbolize random variables and arrows to represent the causal effect between two random variables in the direction of the arrow (~~Cunningham, 2021~~) (Cunningham, 2021; Huntington-Klein, 2021).

~~According to our diagram, there is a direct effect of calculated accessibility on perceived accessibility. We hypothesize that there is an association between objective accessibility, perceived accessibility, and employment outcomes. However, there are other ways in which they are connected, represented by the other nodes of the diagram. First, there are mediator components that influence the way a person perceives calculated accessibility. In our study, we use place-based measures to calculate accessibility, so that these indicators represent neighbourhood characteristics (CT level). However, each individual will perceive these, or make use of, objective accessibility. Each individual perceives the same neighbourhood characteristics differently based on their capabilities and preferences, as well as their travel options and experiences. For instance, a person with a disability that prevents or makes it difficult for them to walk or cycle will probably not enjoy the benefits of active accessibility to the same extent as a person without such a disability (Church and Marston, 2003); someone without a car will not take advantage of high car accessibility; someone without a monthly transit pass might take less public transit than someone with a transit pass, therefore reducing their potential for spatial interaction promoted by higher public transit accessibility.~~

~~In the causal diagram, the accessibility of a neighbourhood is caused by urban characteristics, such as land use and transportation infrastructure. In addition, place of residence, and consequently their an individual's neighbourhood accessibility, is affected (caused) also influenced by socioeconomic factors, with household income as the main driver of this decision, and also by demographic characteristics (such as gender, race, citizenship, among others). These socioeconomic and demographic factors not only determine influence where someone will live, but also shape how individuals experience travel and, consequently, make use of their neighbourhood accessibility.~~

However, ~~it is also worth mentioning that~~ socioeconomic and demographic factors and the urban landscape of the place of residence are also linked through a common cause (U_1) that we cannot measure ~~or properly define as a variable causing both. For instance, directly, such as generational wealth, which likely~~

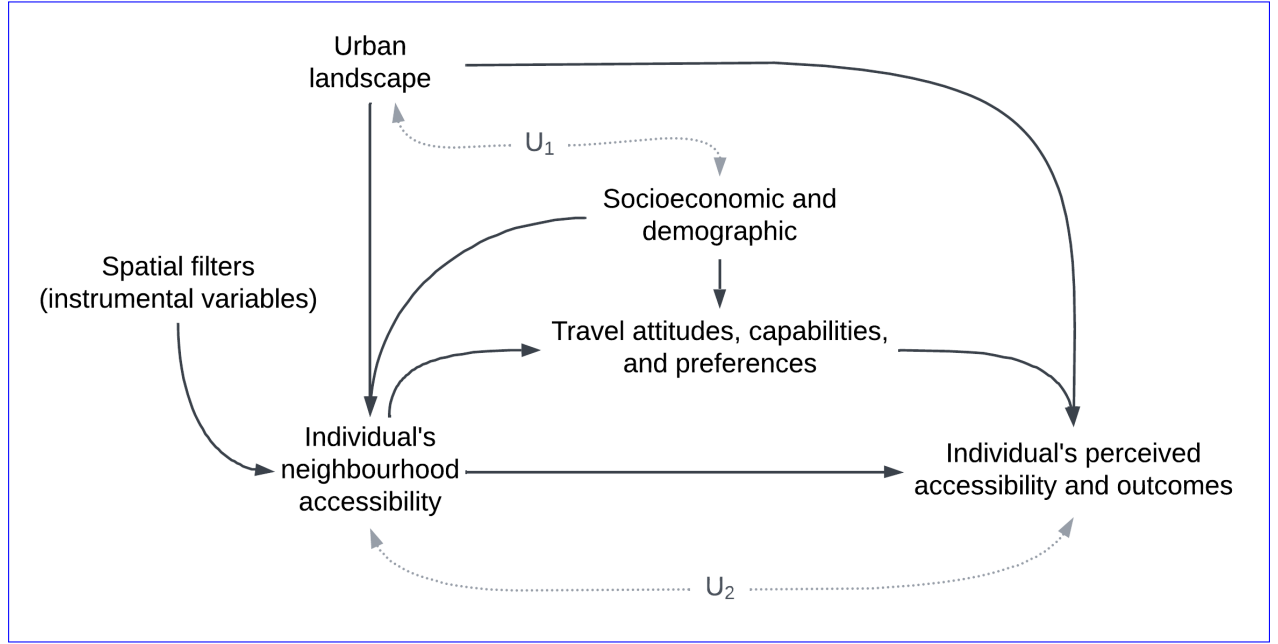


Figure 2: ~~Causal diagram~~ Directed acyclic graph (DAG) of ~~calculated objective~~ accessibility on perceived accessibility.

~~influences both~~ someone's neighbourhood and their income level ~~are probably related to generational wealth,~~ a variable that cannot be obtained from the surveys. In addition, there must exist other factors, unknown or not translated into measurable variables, that probably affect both calculated and perceived accessibility (U_1). Other unmeasured factors, such as residential self-selection and adaptive preference, likely affect objective accessibility, perceived accessibility, and employment outcomes (U_2).

From a causal diagram, we can identify all paths through which we can go from one node (a variable) to another using the arrows connecting them, with particular importance given to those paths connecting the treatment to ~~Because other paths link objective accessibility to perceived accessibility and employment outcomes,~~ the outcome. Paths in which all arrows point away from the treatment are front-door paths (Cunningham, 2021; Huntington-Klein, 2021); whereas paths that point towards the treatment are back-door paths. To identify the effect of the treatment on the outcome, we need to control for at least one variable in the back-door paths (hold this variable constant and unchanged), in a process that closes these biasing paths, while not controlling for at least one front-door path (Huntington-Klein, 2021). In this study, we employ regressions to estimate the relationship between treatment and outcome, ~~observed association between them~~ cannot be assumed to reflect the true effect of accessibility on how people experience and evaluate their access to employment opportunities. If objective accessibility is to be used as a benchmark or standard for equitable transportation planning, we believe it is important to apply methods that aim to isolate its association with perceived accessibility and employment outcomes, after accounting for these confounding mechanisms, instead of relying on a raw statistical association that may be inflated, offset, or masked by unobserved individual preferences and constraints. Our approach relies on, first, controlling for ~~all other~~ measurable variables.

To model the causal diagram from Figure 2, we use weighted regressions that combine multimodal spatial availability per person with variables at the individual and household levels from the NSTP. Considering respondent weights, we transformed the perception variables into an ordered Likert scale based on their quintile position, categorizing them as: "Most Negative", "Negative", "Neutral", "Positive", and "Most Positive". The thresholds used for this categorization is presented in Table 2 ~~individual-level covariates capturing demographic, socioeconomic, and travel-related characteristics; and second, applying a framework~~

using instrumental variables.

~~Quintile thresholds used to categorize outcomes.~~

~~Based on the causal diagram and on our literature review, we selected the following variables to represent each node.~~ Guided by the diagram and the supporting literature, we identified control variables corresponding to each node, following standard causal-diagram identification methods (Cunningham, 2021; Huntington-Klein, 2021). We selected the following variables:

- ~~Calculated Objective~~ accessibility (treatment): multimodal spatial availability per person ~~(active, public transit, and motorized modes)~~ by transportation mode.
- Perceived accessibility (outcome 1): ~~individual perceptions categorized on a Likert scale, obtained from NTSP1~~: agreement with “there are too few jobs available for me in my ideal commuting time,” obtained from NSTP.
- Self-reported employment outcome (outcome 2): agreement with “I have had to decline employment opportunities because of my transport situation,” obtained from NSTP.
- Travel attitudes, capabilities, and ~~preferences~~ preferences: most used transportation mode; whether the preferred transportation mode matches the most frequent mode; whether the individual has a monthly or annual public transit pass; number of cars and bikes in the household; and whether the person has any disability, all obtained from ~~NTSP~~ NSTP.
- Socioeconomic and demographic: whether the person is Indigenous or from a racialized population group; whether the person self-identifies as a woman; or as a transgender or non-binary person; age; household size and household income; and citizenship (temporary visa or refugee), all obtained from ~~NTSP~~ NSTP.
- Urban landscape: population centre class (rural, small, medium, or large urban population centre) and population density (dissemination area level), obtained from Census.

In total, we estimate two regression models, one for each dependent variable: perceived accessibility and declined job offers. Both models use the instrumental variable approach described in the following subsection, and each includes the three accessibility indicators (active, public transit, and private vehicle spatial availability per person), the interactions between active and transit accessibility and their respective user indicators, and the full set of control variables at the individual and household levels from the NSTP.

Although respondents expressed their level of agreement on a continuous scale from 0 to 100, we treat these responses as ordinal measures, not interval measures. Agreement scales based on sliders do not guarantee that equal numerical differences reflect equal differences in perception: a change from 10 to 20 may not represent the same variation in agreement as a change from 80 to 90. Furthermore, respondents may use the scale in different ways, for example, by grouping their responses around the extremes or the midpoint. According to ?, analyzing ordinal responses as if they were metric can distort the estimated effects and inflate error rates. We can, however, have more confidence in the order of the responses, since a respondent who selects 80 agrees more strongly than one who selects 60.

Therefore, using respondent weights, we transformed the dependent variables into an ordered Likert scale based on their quintile position, categorizing them as: “Most Negative”, “Negative”, “Neutral”, “Positive”, and “Most Positive”. The thresholds used for this categorization are presented in Table 2. Since greater agreement with either statement indicates a worse situation, the highest quintile of agreement corresponds to “Most Negative”. Quintiles also ensure a sufficient and balanced number of respondents in each category, which is needed for stable estimation of the ordinal models, and provide a middle category that serves as the neutral reference for the dependent variable.

Considering that the outcome is an ordinal categorical variable, we used weighted ordinal logistic regressions (Lumley, 2011) to model the ~~effect of the calculated accessibility indicators on the perception categories~~ relationship between the objective accessibility indicators and the categories of the dependent variables, using Equation 3:

$$\log\left(\frac{P(Y \leq j)}{P(Y \geq j)} \frac{P(Y \leq j)}{P(Y > j)}\right) = \theta_j - (\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p) \quad (3)$$

Table 2: Quintile thresholds used to categorize dependent variables.

Outcome	20%	40%	60%	80%	100%
Few jobs available in my ideal commuting time	35	52	69	87	100
Declined job offers because of my transport situation	8	26	51	72	100

where Y represents the perception-response category of the respondent (ranging from “Most Negative” to “Most Positive”), and $X_1, X_2, \dots, X_p$ represent the set of explanatory variables, and θ_j and β_p are unknown parameters to be estimated. The parameters β_p are constant for all outcome categories and measure the change in the log-odds of reporting higher perception-response categories associated with each explanatory variable, while θ_j represents the intercept separating the perception-categories-response-categories.

Because of the high correlation between public transit and bicycle accessibility (with Pearson correlation above 0.80), and because only 61 NSTP respondents (less than 3% of the sample) mentioned cycling as their main commuting mode, we combined walking and cycling into a single active transportation indicator, summing the jobs spatially available by each mode. Including bicycle accessibility as a separate indicator, given this small subsample and high collinearity with transit accessibility, would likely have caused convergence problems when creating our ordinal models. Fitting these models also requires a sufficient number of respondents within each category and combination of categories, which the small cycling subsample could not support. However, we acknowledge that this combined active measure does not account for the difference in travel speed between walking and cycling, which can be substantial, and may understate the accessibility gains cycling offers relative to walking over longer distances.

Table 3 presents the variables included in the regression models and their descriptive statistics for the NSTP sample (weighted mean and weighted standard deviation for continuous and dummy variables, and weighted count and percentage for categorical variables, created using the {gtsummary} R package (Sjoberg et al., 2021)).

Weighted summary statistics of the explanatory variables.

3.3.1. Controlling for endogeneity Instrumental variables

The presence of U_2 in our causal diagram-DAG implies the possibility of bias caused by omitted variables , in which unobserved factors simultaneously affect both calculated and perceived accessibility. Omitted variable bias is the that simultaneously affect objective accessibility, perceived accessibility, and employment outcomes - a source of endogeneity , which occurs when an independent variable correlates with the regression error term and that compromises the consistency of the estimators (Antonakis et al., 2010; Cunningham, 2021; Huntington-Klein, 2021). In this study, we address the potential endogeneity by applying We address this using an instrumental variable (IV) technique. An instrumental variable, where an instrument Z must be correlated with the outcome only through objective accessibility. Examples of instrumental variables used in studies analyzing the relationship between employment outcomes and job accessibility include Euclidean distance from the spatial unit centroid to the nearest major road or employment subcenter (Delmelle et al., 2021; Jin and Paulsen, 2018), share of automobile ownership (Hu, 2017), transit accessibility and car unavailability (Johnson et al., 2017), distance to rivers (Duarte et al., 2023; Luz and Portugal, 2022), and population density (Bastiaanssen et al., 2021, 2025).

However, two-Two challenges arise in the search for relevant instruments: i) the likelihood that the instruments and error terms are correlated increases as the correlation between the likelihood of correlation between instruments and the error term increases with the strength of their correlation with the endogenous variable and the instruments becomes stronger, which can render the instruments invalid; and ii) instruments that are, while instruments only weakly correlated with the endogenous variable may be ineffective and tend to perform poorly. In response to these challenges, Le Gallo and Páez (2013) developed an approach that relies on the use of synthetic instruments derived from the spatial configuration of the data (see Getis and Griffith, 2002; Griffith, 2003). Synthetic variables generated via spatial filtering ensure, which ensures high correlation with the treatment and whilewhile, potentially, remaining exogenous to the outcome.-

Table 3: [Weighted summary statistics of the explanatory variables, based on the NSTP sample.](#)

Explanatory variables	N = 3,576,893
Active job accessibility per active traveller	0.06 (0.13)
Transit job accessibility per transit user	0.14 (0.09)
Private vehicle accessibility per user	1.00 (0.27)
Active commuter	528,927 (15%)
Public transit user	829,940 (23%)
Person with disability	716,861 (20%)
Number of Cars	1.30 (0.92)
Number of Bikes	1.15 (1.33)
Preferred mode match most frequent mode	2,706,882 (76%)
Indigenous or racialized person	1,866,043 (52%)
Temporary visa or refugee	268,654 (7.5%)
Woman, transgender or non-binary person	1,757,263 (49%)
Age	40.15 (13.49)
Household size	2.90 (1.61)
Household income	
Less than \$40,000	561,182 (16%)
\$40,000 to \$80,000	1,145,633 (32%)
\$80,000 to \$120,000	930,128 (26%)
More than \$120,000	939,950 (26%)
Population density (people/km2)	7,809.82 (13,610.23)
Population centre class	
Rural	89,251 (2.5%)
Small	48,306 (1.4%)
Medium	58,128 (1.6%)
Large	3,381,208 (95%)
Public transit availability in the neighbourhood	3,435,002 (96%)
Monthly or annual public transit pass	1,063,225 (30%)

¹ Mean (SD); n (%)

~~Following the methodological approach of Le Gallo and Páez (2013) to create spatial filters for the accessibility variables~~outcome. Following this approach:

1. We built a queen-contiguity matrix for the census tracts to represent spatial weights, expressing the existence of a neighbouring relation as a binary relationship (1 for neighbours and 0 otherwise).
2. Then, we computed all eigenvectors associated with the contiguity matrix. These eigenvectors are orthogonal to one another and represent a portion of the variance in the spatial contiguity matrix.
3. After this, we initialized a variable S with value 0 and linearly regressed each accessibility indicator (the independent variable) on each eigenvector and S . If the eigenvector coefficient was significant (adopting a p -value ≤ 0.05), we multiplied the eigenvector by its coefficient and added it to the value of S . After this process, S represents the spatial filter associated with the accessibility indicator.

All spatial filters presented a Pearson correlation above 0.88 with their respective accessibility measures and absolute correlations lower than 0.15 with the original ~~perception-continuous~~ dependent variables (before transforming them into categorical variables).~~These results suggest that the spatial filters are strongly correlated with the endogenous accessibility variables while remaining weakly associated with the outcome, which is~~, consistent with the relevance and exclusion conditions required for valid ~~instrumental variables~~instruments. In addition, we performed ~~the Wald test~~Wald tests to evaluate the null hypothesis of weak instruments and insufficient explanatory power to predict ~~our endogenous variables~~objective accessibility, and Wu-Hausman tests to assess whether the accessibility indicators are endogenous. We report both diagnostics for the categorical models and, as a robustness check, for linear models estimated on the original continuous dependent variables (Appendix A.2).

Since our dependent variables are not linear, the popular technique of replacing endogenous variables by their first-stage predictors (two-stage least ~~square-squares~~ estimation) is not appropriate - the so-called “forbidden regression” in econometrics (Wooldridge, 2010; Cunningham, 2021; Huntington-Klein, 2021). Therefore, we applied the two-stage residual inclusion (2SRI) approach (Terza et al., 2008), in which~~in the first-stage~~in the first stage, we estimated the accessibility ~~measures-indicators~~ using a linear regression including all control variables plus spatial filters (our instrumental variables). Next, in the second-stage, the dependent variables were estimated using the accessibility measures, the control variables, and the residuals from the first-stage regressions. ~~In both stages, the respondent weights were considered to perform the regressions~~Both stages used respondent weights. In this approach, we use the regular treatment to predict the outcome, but also control for the unexplained part of the treatment.

3.4. ~~Equity Standards~~Identifying equity standards

Considering only the cases in which the ~~calculated-objective~~ accessibility was statistically significant, we measured changes in the predicted probabilities across ~~perception-response~~ categories by varying the levels of ~~calculated-objective~~ accessibility, while controlling for the other explanatory variables. There are many possible ways of controlling the explanatory variables (e.g., fixing the number of cars in the household as 0, 1, or 2+; or fixing the respondent age as 30, or 31, and so on). Based on this, we decided to evaluate the effects of ~~calculated-objective~~ accessibility in three distinct individual profiles:

- ~~An equity-deserving~~A transport-disadvantaged profile, with individual-level control variables set to the values corresponding to populations facing the greatest transportation challenges.
- The most ~~common-prevalent~~ profile, reflecting modal characteristics of the study area population and with control variables set to their most common values; and,
- An advantaged profile, defined by selecting for the most privileged population segment in terms of their transport experience.

For each individual profile, we increased the value of each accessibility measure by 0.01 jobs per person, while holding constant other accessibility measures (representing alternative transportation modes) at their weighted median values. We also adopted the weighted median value for the population density and a

constant zero value for the residuals obtained from the first-stage regressions. For each transportation mode, we evaluated the effect of accessibility by considering the individual profiles as users of that specific mode. That is, for active travel accessibility, all individual profiles were set as active travellers, and so on for the remaining modes.

Then, for all individual profiles, we identified the minimum value of ~~calculated job accessibility per person in which the objective job accessibility at which the combined~~ probability of the positive and neutral perceptions combined was higher than the probability of being in negative perceptions, naming this minimum value as a non-negative response categories (“Neutral”, “Positive”, and “Most positive”) exceeds the combined probability of the negative categories (“Negative” and “Most negative”). Because the five probabilities sum to 100%, this is equivalent to the probability of a negative response falling below 50%. We name this value the conservative threshold. Next, we defined a the progressive threshold as the minimum accessibility value that resulted in the probability of positive perceptions (not considering neutral perceptions) being higher than of negative perceptions. We then identified an intermediate threshold, set at which the combined probability of the positive categories (“Positive” and “Most positive”) exceeds the combined probability of the negative categories, excluding the neutral category from both sides. Finally, we set the intermediate threshold as the midpoint between the conservative and progressive thresholds.

~~Then, we selected as the equity standards the accessibility thresholds obtained. We base our methodology on egalitarian and sufficientarian perspectives, and we propose defining equity standards based on the equity-deserving individual profile. The rationale behind this equity standard definition is to identify transport-disadvantaged profile. Our approach to operationalizing these justice principles relies on identifying the minimum accessibility level that gives (sufficientarianism) at which the most disadvantaged population a higher probability of having at least a neutral perception profile (egalitarianism) is more likely to give a non-negative response than a negative one (conservative standard), a neutral to positive perception (intermediate standard), or a positive perception and more likely to give a positive response than a negative one (progressive standard).~~

3.5. Measuring transportation equity

We calculated the extent and severity of accessibility ~~gap-gaps~~ using FGT measures (Foster et al., 2010), according to the Equation 4. FGT measures ~~evaluates evaluate~~ the number of people below a standard and the size of the accessibility ~~shortfall-gap~~ relative to it.

$$FGT_{\alpha} = \frac{1}{N} \sum_H^{i=1} \left(\frac{z - y_i}{z} \right)^{\alpha} \quad (4)$$

Where N is the size of the total population; z is the accessibility standard; H is the number of individuals below the z ; y_i is the accessibility level of each individual i ; and α : determines how sensitive the index is to ~~change-changes~~ in the accessibility gap.

Changing the values of α enables the calculation of different FGT indices (Karner et al., 2024). When $\alpha = 0$, FGT_0 measures the extent of undersupply as a simple headcount, indicating the proportion of people below the standard. When $\alpha = 1$, FGT_1 measures the severity of the undersupply as the average percent distance between the standard and the accessibility levels of the individuals below it.

3.6. Areas for priority intervention

~~For each equity standard, we identified CTs in the accessibility was lower than the equity threshold. Next, we calculated the AFI (Martens, 2016b) to measure the severity of the insufficiency, by multiplying the population size below the standard by the accessibility gap. In practice, the AFI is a simple way to highlight areas with large groups of people that fall the furthest behind the set standards. After this, we classified CTs with accessibility shortfall into levels of priority for policy interventions: Very-high (AFI above the third quartile, top 25%), High (AFI between the median and third quartile, 25-50% from the top), Moderate (AFI between the first quartile and the median, 50-75% from top), and Low priority (AFI below the first quartile, bottom 25%). We applied the FGT measures to the general population in each CT, and also to the population classified as low-income according to the Census, as defined by the LIM-AT.~~

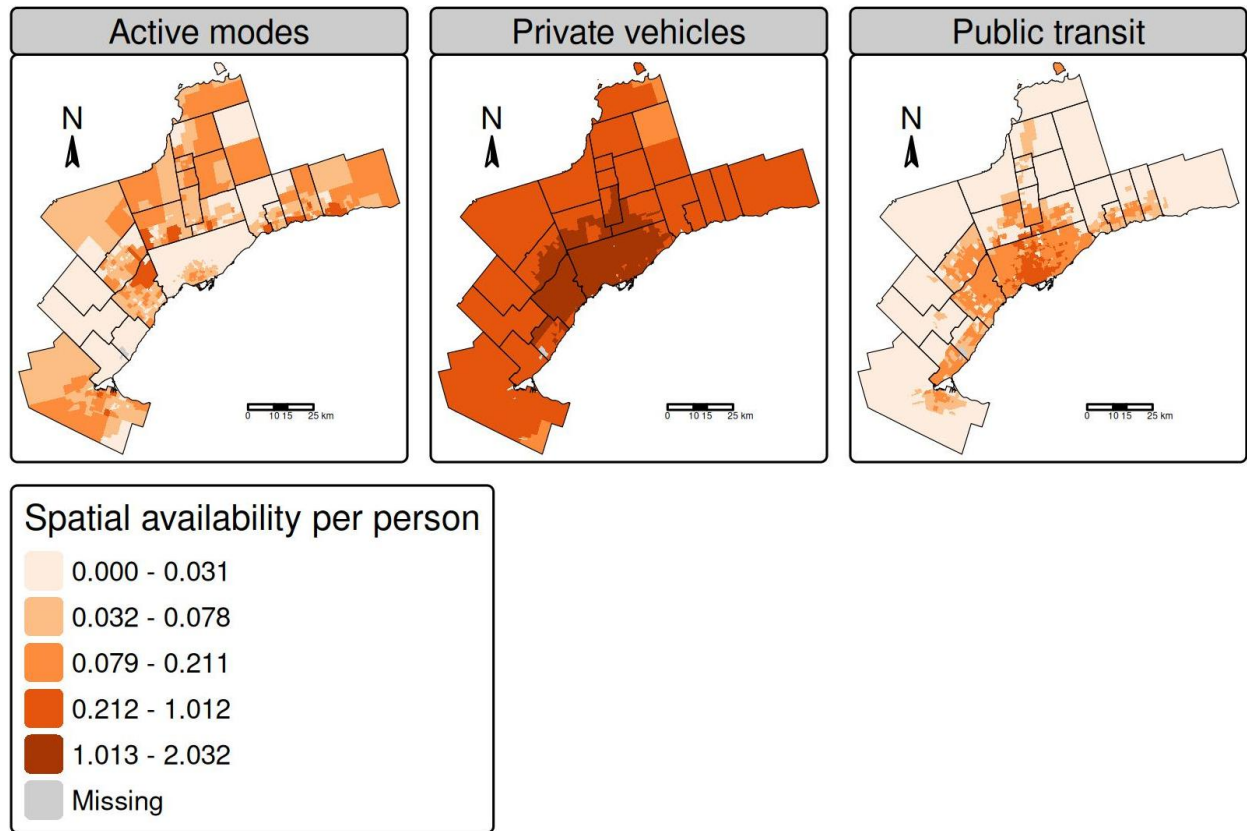


Figure 3: Maps of the multimodal spatial availability per person, considering active, public transit and ~~motorized modes~~private vehicles (accessibility values are classified using quintile breaks).

4. Results and discussions

4.1. Multimodal spatial availability per person.

Figure 3 presents the maps with the multimodal spatial availability of jobs per person. These maps show how, generally, ~~motorized mode~~private vehicle users can access more job opportunities, but also show that there is not much spatial variation when compared to the other two modes, with most of the CTs in the study area classified in the second-highest ~~and~~ or highest accessibility categories. Toronto and Mississauga present the highest level of ~~motorized~~private vehicle accessibility.

Transit accessibility appears to be more spatially distributed across the study area, with the city of Toronto presenting the highest rate of jobs accessible per transit commuters. Overall, we can observe an expected spatial pattern where CTs close to public transit infrastructure present higher levels of transit accessibility when compared to suburbs and rural areas, which often lack public transit services.

Next, the map of the active jobs accessible per active traveller (walkers or cyclists) does not present a clear spatial pattern, with many suburban and rural areas presenting higher levels when compared to more urbanized areas. This can be explained by either a higher number of jobs accessible within typical walking or cycling trip durations, less probable; or, probably, because the number of active travellers in these areas is very low, resulting in a higher rate of jobs per person, in a way that higher accessibility values in these areas may reflect low competition rather than higher absolute job availability.

Although ~~motorized~~private vehicle commuters represent about 78% of the labour force, they compete for 96% of the jobs in the study area. Transit users represent 15% of the labour force, but they reach only

3.4% of the jobs available, while the remaining 0.6% ~~percent~~ of jobs available are competed for by the 7% of active users. On average, each active user respondent from the NSTP can access 0.06 jobs, while each transit user can access, on average, 0.14 jobs. ~~These both~~ Both these measures are substantially lower than the number of jobs that each ~~motorized private vehicle~~ commuter can reach, which is approximately one job per commuter on average.

4.2. ~~The effect impact of calculated accessibility on perceived objective~~ accessibility

Table 4 presents the two probability models linking ~~calculated objective~~ accessibility to perceived accessibility ~~and employment outcome~~. Based on Wald tests, we reject the null hypothesis of weak instruments and insufficient explanatory power to predict our endogenous variables (calculated the objective accessibility indicators) based on Wald tests, given that all of the, as all F-statistics are much higher than well above the Stock-Yogo critical values and are statistically significant (p-value < 0.05). However In the categorical models, the first-stage residuals were non-significant in the second-stage regressions, suggesting ~~that there is~~ little evidence of endogeneity in the ~~calculated objective~~ accessibility indicators, and that standard regressions ~~(without the IV approach)~~ could also have been appropriate. ~~Despite that, we decided to continue our analysis using the models that account for potential endogeneity~~ However, a Wu-Hausman test on the linear (continuous) version of the employment-outcome model does detect endogeneity (p < 0.05; Appendix A.2), suggesting that the categorical model's failure to detect it may reflect a loss of statistical power from quintile-categorization rather than genuine exogeneity. Given this, we retained the IV framework throughout our analysis rather than reverting to standard regressions.

~~Calculated Objective~~ accessibility indicators were partially significant across the models. ~~Motorized Private vehicle~~ accessibility did not present significant values for any of the models, probably due to its ubiquity and homogeneity across the study area.

The model related to the perception of job availability (i.e., “there are too few jobs available for me in my ideal commuting time”) ~~captures well the expected behaviour of accessibility produces an estimated association consistent with our hypothesized causal relationship~~, in which increases in ~~calculated objective~~ transit and active accessibility ~~is are~~ associated with a higher probability of having a more positive ~~perception~~ of accessibility. In fact, this is the model in which the ~~outcome dependent variable~~ is the most similar to the spatial availability of jobs definition, which refers directly to the number of jobs accessible, calculated using the impedance functions calibrated based on population behaviour. For this model, ~~one additional job accessible by transit a one-standard-deviation increase in transit accessibility (about 0.09 jobs per transit user increases the average) is associated with 25% higher odds of being in a more positive perception group by 25% response category~~, indicating that fewer respondents agree that there are too few jobs accessible within their ideal commuting time. ~~Active accessibility plays an overall significant effect for Among non-active travellers, reducing the a one-standard-deviation increase in active accessibility (about 0.13 jobs per active traveller) is associated with 14% lower odds of being in a more positive perception category by 14% response category~~. For active travellers, on the other hand, the ~~interaction effect positive interaction term~~ more than offsets this ~~to result in a net positive effect, so the combined association is slightly positive~~. The same occurs in the model related to ~~declining employment declined job offers due to transport constraints~~, whereby active accessibility is ~~negative negatively associated with outcomes~~ for non-active travellers, but ~~positive positively associated~~ for active travellers. We believe that the ~~general negative effect of active accessibility for negative association between active accessibility and responses among~~ non-active travellers can be explained in terms of job competition, since increasing the number of jobs accessible for a mode that an individual does not use implies greater competition for jobs accessible through other modes, effectively reducing perceived accessibility for the mode a person actually uses. In addition, as previously shown in Figure 3, many of the areas with high active accessibility are located outside downtown cores, in more suburban areas with a low concentration of job opportunities. This ~~further~~ reinforces the assumption that some suburban areas exhibit high active accessibility not because of greater job availability, but due to a very low number of active travellers, resulting in a higher rate of jobs per person - reflecting low competition for jobs accessible by walking and cycling rather than higher active job availability.

Table 5 presents the control variable values applied for each individual profile, while Figure 4 displays the specific effect of increasing transit accessibility across the selected individual profiles for the “few jobs

Table 4: Weighted ordinal logistic regression models connecting ~~calculated objective~~ to perceived accessibility and to employment outcome.

Variable	Few jobs			Declined job offers		
	OR	SE	p-value	OR	SE	p-value
Active job accessibility per active traveller	0.86*	0.064	0.024	0.84*	0.085	0.033
Active mode interaction: accessibility $\times$ user	1.17*	0.072	0.029	1.21*	0.091	0.036
Transit job accessibility per transit user	1.25*	0.112	0.044	0.89	0.191	0.5
Transit mode interaction: accessibility $\times$ user	0.95	0.133	0.7	1.16	0.208	0.5
Private vehicle accessibility per user	0.93	0.097	0.4	1.17	0.147	0.3
Number of Cars	0.98	0.121	0.9	1.11	0.137	0.4
Number of Bikes	1.08	0.067	0.2	1.02	0.066	0.7
Preferred mode match most frequent mode	1.14	0.168	0.4	1.27	0.246	0.3
Monthly or annual public transit pass	0.68*	0.174	0.026	0.52*	0.257	0.012
Public transit availability in the neighbourhood	0.75	0.245	0.3	1.04	0.228	0.9
Indigenous or racialized person	0.60**	0.164	0.002	0.57**	0.211	0.008
Temporary visa or refugee	1.61	0.257	0.064	0.70	0.421	0.4
Person with disability	0.69	0.209	0.077	0.80	0.232	0.3
Woman, transgender or non-binary person	1.05	0.149	0.7	1.02	0.195	>0.9
Age	1.01	0.006	0.080	1.02*	0.007	0.023
Household size	1.06	0.052	0.2	0.94	0.068	0.4
Household income			0.11	**		0.004
Less than \$40,000	—	—		—	—	
\$40,000 to \$80,000	0.80	0.188		1.50	0.272	
\$80,000 to \$120,000	1.23	0.220		1.99	0.250	
More than \$120,000	1.29	0.227		3.08	0.314	
Population density (people/km2)	1.18*	0.075	0.024	1.10	0.101	0.3
Population centre class			0.3	***		<0.001
Rural	—	—		—	—	
Small	1.34	0.487		0.86	0.484	
Medium	2.35	0.469		5.13	0.457	
Large	1.25	0.322		1.16	0.331	
Residuals: Active accessibility	1.09	0.078	0.3	0.97	0.085	0.7
Residuals: Transit accessibility	0.85	0.092	0.073	0.85	0.148	0.3
Residuals: Private vehicle accessibility	1.13	0.068	0.075	0.97	0.094	0.7
Wald F	2.86			4.85		
p-value	<0.001			<0.001		
AIC	6,829			4,312		

Abbreviations: OR = Odds Ratio, SE = Standard Error

*p<0.05; **p<0.01; ***p<0.001.

Accessibility indicators, population density, and first-stage residuals are standardized; odds ratios correspond to a one-standard-deviation increase.

Table 5: Individual profiles to estimate probabilities.

to >X>X>X>X	Variable	Equity-deserving	Most common population	Advantaged group	Number of Cars	0	1	2	3
1-2	Number of Bikes	0	1	1	Preferred mode match most frequent mode	No	Yes	Yes	Indigenous or racialized person
		Yes	Yes	No	Temporary visa or refugee	Yes	No	No	Person with disability
No	Woman, transgender or non-binary person	Yes	No	No	Age	35	40	45	Household size
		3	3	3	Household income	Less than \$40,000	\$40,000 to \$80,000	More than \$120,000	Population centre class
		Small	Large	Large	Public transit availability in the neighbourhood	Yes	Yes	Yes	Monthly or annual public transit pass
No		Yes	No	No		Yes	No	No	

Bold variables are statistically significant (p-value < 0.05) for at least one of the models.

available” model. For all of them, we observe a decrease in negative perceptions as ~~calculated-objective~~ transit accessibility increases. For the ~~equity-deserving-transport-disadvantaged~~ individual profile, the “Most negative” perception has a high probability at low levels of ~~calculated-objective~~ accessibility. In fact, at the median ~~calculated-objective~~ transit accessibility (0.12 jobs accessible by public transit per transit user, calculated based on all NSTP respondents), the two most negative perception categories (“Most negative” and “Negative”) together account for over 52% of the total probability, resulting in a high likelihood that individuals in the ~~equity-deserving-group-to-transport-disadvantaged-group-will~~ have negative perceptions at this accessibility level. For the most ~~common-prevalent~~ population, the initial pattern appears similar to the ~~equity-deserving-transport-disadvantaged~~ group, but the decline in negative perceptions occurs faster, with the ~~combined probability of positive responses (“Positive” and “Most positive” probability surpassing negative probabilities) surpassing that of negative responses (“Negative” and “Most negative”)~~ at approximately 0.28 jobs per transit user. In contrast, for the advantaged population, we can see that even at very low levels of accessibility, the positive perceptions already present the highest probabilities and increasing transit accessibility only reinforces these positive perceptions.

Regarding the control variables, in the model related to “there are too few jobs available for me in my ideal commuting time”, having a monthly or annual public transit pass is associated with a 32% ~~reduction in the lower~~ odds of being in a more positive category. Being an Indigenous or racialized person is associated with a reduction of 40% in the odds of more positive perceptions, while ~~increasing one person per square kilometres (population density a one-standard-deviation increase in population density (13.610 people/km))~~ was associated with an 18% ~~increase in the odds of being in a more positive category higher odds~~.

In the model related to “I have had to decline employment opportunities because of my transport situation”, having a monthly or annual public transit pass ~~reduced the was associated with 48% lower odds~~ of being in a more positive category by 48%; and being Indigenous or racialized ~~reduced the odds by with 43% lower odds~~; and each additional year of age is associated with an increase of 2% in the odds of being in a more positive group. In terms of income, higher income is associated with more positive ~~perceptions (less perceived employment decline outcome (fewer reports of declined job offers))~~, with households earning more than \$120,000 CAD having approximately three times higher odds of being in a positive category

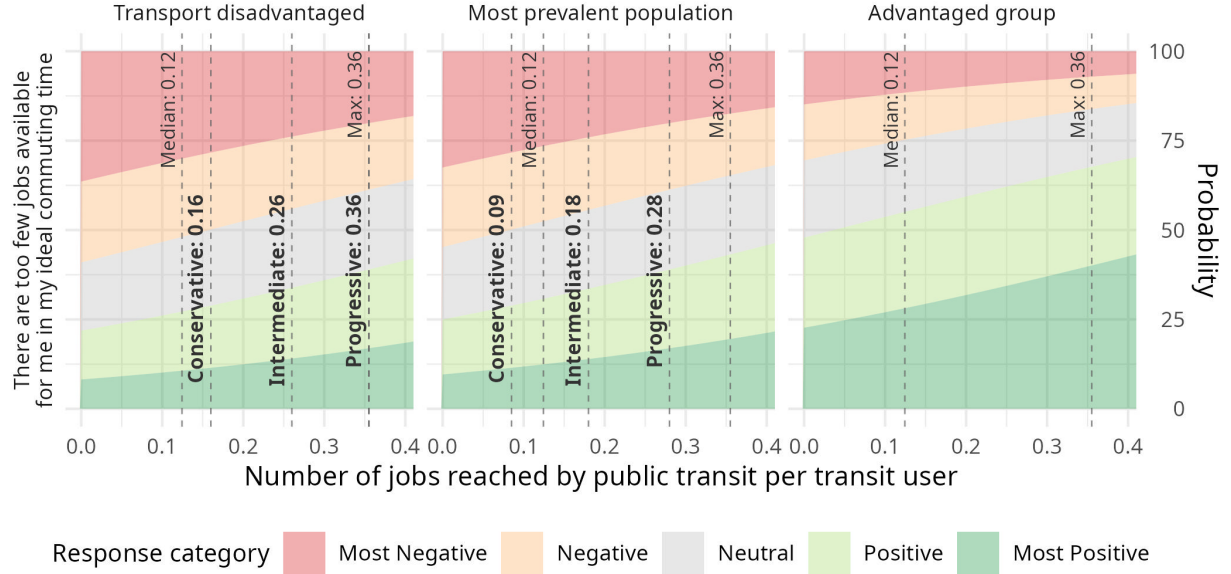


Figure 4: Public transit accessibility thresholds based on selected individual profiles.

compared to those earning less than \$40,000 CAD. In relation to the census population centre class, living in a medium-sized population centre is associated with substantially more positive ~~perceptions-outcomes~~ (over five times higher odds), while living in a small centre is associated with ~~12~~14% lower odds, and living in a large centre is associated with 16% higher odds - all this when compared to rural areas.

4.3. Equity standards and their impact on ~~transportation-transportation~~ equity

In this section we use the models presented above to identify active accessibility thresholds, while only the “there are too few jobs available for me in my ideal commuting time” model was selected to identify ~~public~~ transit thresholds - since ~~e-calculated-objective~~ transit accessibility was statistically significant only in this model.

Figure 4 shows the conservative, intermediate, and progressive public transit thresholds calculated for each individual profile. For the advantaged profile, a level of 0 jobs per transit user is already enough to result in positive perceptions, which means that the current transit accessibility levels would be sufficient and no individuals would fall below the minimum accessibility if this profile was selected as the reference. However, it is worth noting that this result should not be interpreted as evidence that public transportation accessibility is irrelevant to the advantaged group’s ability to reach employment. Instead, it demonstrates that the predicted probability of a positive perception for this profile is determined primarily by its other characteristics, such as higher household income and not being a racialized person, so that variation in accessibility has little additional effect. This is consistent with our DAG, which shows that individual resources mediate the relationship between objective accessibility, perceived accessibility, and employment outcomes, such that the more advantaged group has a greater ability to overcome low spatial accessibility through other means.

For the most ~~common-prevalent~~ profile, the conservative standard (the minimum level of accessibility at which the sum of negative perceptions is lower than the sum of all other perceptions) is equal to 0.09 jobs per transit user. This threshold is lower than the median ~~calculated-objective~~ transit accessibility. The progressive threshold, defined as the minimum level at which the sum of positive perceptions exceeds the sum of negative perceptions, for this profile is 0.28 jobs per transit user, with the intermediate threshold defined as the midpoint between these two values (0.18). However, for the ~~equity-deserving-transport-disadvantaged~~

profile, the conservative threshold is equal to 0.16 jobs per transit user, which is higher than the median accessibility level. The progressive standard is equal to 0.36 jobs per transit user, which corresponds to the maximum observed level of accessibility in the study area, and the intermediate threshold is 0.26.

Evaluating the thresholds for active accessibility, we again find that even a minimal level of accessible jobs per active traveller is already ~~enough to result in positive perceptions~~ related to positive perceptions and outcomes for the most advantaged profile ~~in both models, meaning that the current active accessibility levels would be sufficient and no individuals would fall below the minimum accessibility threshold~~. For the most ~~common-prevalent~~ profile, the identified conservative threshold is also minimal; ~~and the the intermediate and progressive thresholds are approximately 1.09 and 2.19 jobs per active user, resulting in higher probabilities of positive perceptions for the models of declining employment and few jobs available, respectively~~ traveller in the declined job offers model, and 1.69 and 3.39 in the few jobs model.

However, for the ~~model related to declining employment opportunities due to transportation conditions~~ declined job offers model, we were not able to identify a threshold ~~that resulted in a higher likelihood of positive perceptions for the equity-deserving at which favourable outcomes became more likely than unfavourable ones for the transport-disadvantaged~~ profile, even after estimating probabilities at more than double the maximum observed active accessibility level (2.03 jobs per active user). For the model regarding the perception that there are too few jobs available within an ideal commuting time, we identified 0.9 as the conservative threshold (22 times the observed median level) and a progressive threshold close to 5 jobs per active user (almost 2.5 times the maximum observed level and 123 times the median). These high thresholds ~~are explained by the relatively small net positive effect of active accessibility, while reflect the small net association between active accessibility and the outcomes, compared with the larger associations of the other explanatory variables exerted a greater influence on with~~ the estimated probabilities ~~of belonging to a more positive perception group~~. Based on both models, the high active accessibility thresholds help illustrate the limitations of potential policies focused ~~solely only~~ on increasing active accessibility to promote ~~positive perceptions~~ a more positive perceived accessibility and employment outcomes: an approach that ~~intervene-intervenes~~ only in the spatial dimension of accessibility, without considering the capacities and travel preferences of the population, ~~would likely result in limited impact~~ may have a limited effect on residents' ~~perceptions regarding employment reachability and related outcomes~~ perceived accessibility and employment outcome.

These thresholds show how, depending on how the socioeconomic and transportation parameters are controlled and how the individual profiles are defined, different sufficient levels of accessibility are obtained. In fact, if one chooses to define thresholds based on the advantaged profile, it might lead to the conclusion that the current accessibility state is already sufficient and that no intervention or improvement is required. Alternatively, ~~equity-deserving transport-disadvantaged~~ groups might still hold negative perceptions depending on ~~whether~~ the threshold based on the most ~~common-prevalent~~ population is selected. Therefore, the only way to ensure that ~~equity-deserving transport-disadvantaged~~ groups have a better chance of ~~experience positive perceptions~~ experiencing positive perceptions and employment outcomes is to define minimum accessibility levels based on the controls tailored to these populations, setting socioeconomic and transportation parameters that represent them, thereby adopting a truly equity-based standard.

Considering the feasibility for policy interventions, we suggest the transit ~~threshold~~ thresholds defined in terms of the ~~equity-deserving transport-disadvantaged~~ profile as examples of equity standards. Around 66% of the study area population lives in areas with ~~lower~~ transit accessibility levels lower than the conservative equity standard (0.16), while 91% live in areas with transit accessibility levels lower than 0.26 jobs accessible by transit per transit user (intermediate equity standard), and almost the entire study area population lives in areas below the progressive equity standard (0.36, which is also the maximum transit accessibility observed in the study area). Regarding the low-income population, 55% live in areas with an accessibility gap under the conservative standard, 87% under the intermediate standard, and almost all live in areas with an accessibility gap under the most progressive standard.

In terms of the severity of transport disadvantage (FGT-1), the average ~~shortfall~~ gap among deprived individuals amounts to 36% under the conservative standard, meaning that those who do not meet the threshold fall short of it by roughly one-third on average (about 0.06). For the low-income population, the average shortfall is around 27% (about 0.04). For the intermediate standard, the average ~~gap~~ shortfall for the



general population is 53% (about 0.14), while it is 44% (0.11) for the low-income group. For the progressive equity standard, the average shortfall for the general population is about 0.23 (65% of the standard), and about 0.21 (58%) for the low-income group. These FGT measures show that, as the minimum transit accessibility levels become more demanding when moving from conservative to progressive standards, not only do more individuals fall below the threshold, but those in accessibility shortfall-gap are, on average, substantially further from achieving it.

4.4. Areas for priority intervention

They also show that the low-income population is, in general, located in areas with greater public transportation accessibility than the general population, meaning that low-income individuals who fall below a certain threshold are, on average, closer to reaching it and would need comparatively smaller improvements in accessibility to reach it.

Figure 6 shows the priority areas for policy intervention for CTs with transit accessibility shortfall, based on their AFI index and for ?? shows the public transit accessibility gap in each CT, calculated as the difference between the CT's transit accessibility level and each equity standard type. All census subdivisions contain CTs experiencing transit accessibility shortfall, independently with a transit accessibility gap, regardless of the type of standard. For the conservative standard, about 32 only 35% of the CTs (431-516 out of the 1,489 CTs in the study area) present transit accessibility levels above the shortfall-limit standard, with most of them (9183%) located in the City of Toronto. This share of CTs without an accessibility gap decreases to less than 10% for the intermediate equity standard (136 CTs). Only one CT, located in Toronto, does not present an accessibility deficit-gap when evaluated under the progressive standard. The unique CT without accessibility shortfall-This unique CT corresponds to the area surrounding the St. George Subway Station in Downtown Toronto, with a connection connected to the two subway lines that cross the city and connect one of which extends into neighbouring municipalities, running from North to South and from East to West. In addition, this north to south and from east to west. This CT is bounded by Spadina Avenue on to the west and Bloor Street on to the south, with many connections to the public transit system, including a streetcar line on Spadina.

In terms of areas for policy intervention, considering absolute population living in accessibility insufficiency when compared to the conservative standard, the subdivision of Brampton is the one with City of Toronto has the largest population living in areas classified as very high priority (217,253 people, representing 33% of the subdivision population in this situation (over 736,000 people, 26% of its residents), followed by Hamilton with 184,991 people (32Brampton (over 635,000 people, 97%), and Hamilton (almost 570,000, 100%). For the low-income population, Toronto again ranks first (almost 86,000 individuals, 24% of its low-income residents), Vaughan (144,670 people, 44%) and Markham (142,066 people, 41) followed by Hamilton (over 60,000, 100%), and Mississauga (over 45,000, 68%). For the most progressive standard, Brampton and Hamilton again appear having the largest populations living in very high priority areas, with 380,289 people (57%) and 229,549 people (40%) respectively, but now followed by the City of Toronto with 221,768 people (7%).

However, in terms of population share, 100% of the population living in the subdivisions of East Gwillimbury (34,637 people), Uxbridge (21,556 people), and Whitechurch-Stouffville (49,864 people) are in areas with very high priority for policy intervention for the conservative and intermediate standards. For the progressive standard, only Whitechurch-Stouffville has all its CTs classified as very high priority areas. This difference occurs because the priority areas are defined based on their AFI position in the quantile classification for the entire study area, and as additional CTs are included under more rigorous standards, the AFI distribution also shifts. Regarding the Whitechurch-Stouffville, this subdivision is served by only one fixed public transit line managed by GO Transit, a regional train line that connects Stouffville with the subdivisions of Markham and Toronto. The list with the total number of people living in each priority area, by equity standard type and subdivision, is available as supplementary material. We believe that the areas with accessibility gaps identified here can help guide future investments in public transportation services and infrastructure. Although suburban areas such as Vaughan and Markham exhibit large absolute accessibility gaps, viewing these differences with a vertical equity perspective suggests different priorities. If focus is on the absolute number and proportion of low-income residents living in areas with an accessibility gap,

Priority areas for policy intervention by type of equity standard.

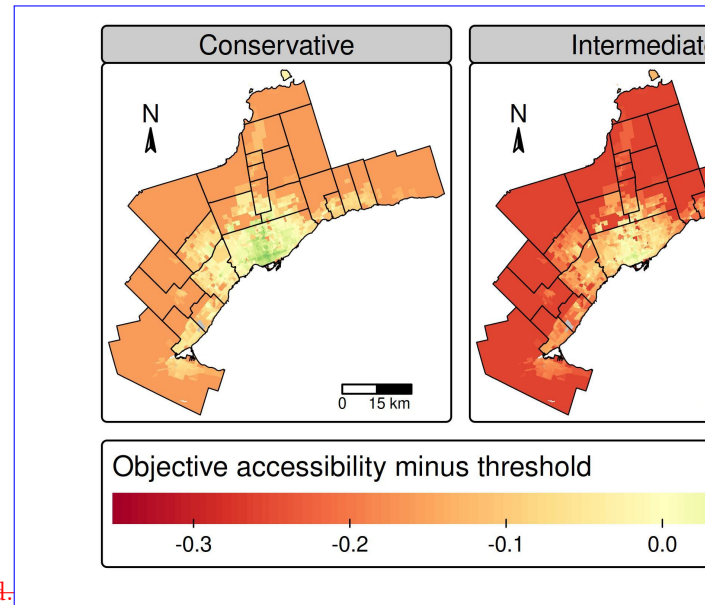


Figure 6: Difference between objective accessibility and thresholds.

Hamilton and Toronto are candidates for intervention. Toronto has the largest low-income population below the standard, while in Hamilton all low-income residents fall below it.

Because objective accessibility depends on transport and land-use factors, gaps can be reduced through two broad types of intervention. First, improvements in transit service aiming to increase public transit accessibility, such as higher service frequencies, new routes, or rapid transit lines, reduce travel times and expand the number of jobs reachable within typical commuting times. Second, land-use policies that bring employment closer to transit, such as transit-oriented development and mixed-use zoning, can increase the number of opportunities reachable without changes to the network.

For instance, the City of Hamilton is currently implementing its first Light Rail Transit (LRT) line in Downtown Hamilton, connecting the eastern and western parts of the city while also crossing Downtown, an area that concentrates a large share of the city’s available jobs. Provided the city mitigates potential displacement and relocation of its low-income population resulting from land price increases associated with the LRT, its inclusion is likely to have a significant impact on transportation equity, since the LRT crosses the areas with accessibility gaps and the largest concentration of low-income residents in the city.

5. Critical analysis and limitations

~~Although the unprecedented look at transport poverty in Canada provided by the NSTP~~

In this study, we present a model-based approach to defining equity standards that achieve a desired normative outcome, based on the estimated relationship between objective accessibility, perceived accessibility, and an employment outcome. We consider this approach our main contribution to transportation research, and more specifically to the literature on the equitable distribution of transportation benefits. Because the resulting standards are based on how residents actually experience accessibility, they are also easier to communicate to policy practitioners and the general public.

Defining minimum levels of accessibility necessarily requires selecting a threshold at which objective accessibility becomes “sufficient” (Martens, 2016b). Following Pereira and Karner (2021), we combine egalitarian and sufficientarian perspectives of distributive justice and define this point as the minimum level of accessibility (sufficientarianism) at which a transport-disadvantaged profile (egalitarianism) is more likely to give a

non-negative response than a negative one (conservative standard), or a positive response than a negative one (progressive standard). In our view, it is important to make explicit the moral foundations that support this choice, because different theories of justice lead to different perspectives on how policies should manage the distribution of transport-related benefits.

However, these standards could have been defined in many other ways, both in terms of the theory of justice guiding their definition and in terms of the threshold considered sufficient. This choice is ultimately normative, and it should not rest solely with researchers. As stated by Karner et al. (2020), transport equity is only one step towards achieving transport justice: a condition in which no person or group is disadvantaged by a lack of access to the opportunities they need to lead a dignified and healthy life. Achieving this condition requires transforming the structures and processes that lead to the unequal distribution of the multiple externalities of transport across population groups and space.

Consequently, the minimum levels of accessibility we propose are intended to guide transportation planning towards a more equitable distribution of transportation benefits, as an initial or an intermediate stage to be followed by a participatory and democratic process to collectively select the most appropriate standard. Residents and other stakeholders must be able to actively participate in, and influence, the decisions that affect their lives. Such a process could define not only the final standard but also involve the public throughout the entire modelling process. This could include deciding which variables should be included in the models, which profile should serve as the reference for defining thresholds, and how the thresholds themselves should be defined. Although we consider this the most appropriate way to define standards, it is beyond the scope of this study to develop democratic plenaries to discuss the suggested standards.

In addition, the scope of our standards is limited, and we do not propose them as fixed or universal benchmarks applicable to other regions. By definition, objective accessibility depends on the study area's land-use and transport characteristics and on the measure selected to represent it, and we would not expect or intend for our thresholds to be transferred elsewhere. However, while our results are context-dependent, we believe that our modelling approach is not.

In terms of the opportunity type, we selected job accessibility as our indicator to define equity standards for a few reasons. First, employment accessibility is widely recognized as a key indicator of how well a city or region serves its residents (Bania et al., 2008; Owen and Levinson, 2015; Shen, 1998), and as an indicator of the “quality of urban living” (Wachs and Kumagai, 1973; Handy, 2020), since employment provides the financial ability to support other life activities (Merlin and Hu, 2017). Second, calculating accessibility requires the use of impedance functions, and we decided to use calibrated functions based on empirical data to better represent people's travel behaviour. Accessibility measures also require land-use data on both population (for competitive measures) and the number of opportunities. Both requirements are met by Census microdata specifically for employment and the labour force, which is not the case for other types of opportunity. Third, the NSTP survey includes questions on perceived job accessibility and employment outcomes specifically, which directly enabled our approach of linking objective job accessibility to individuals' perceptions and outcomes related to this opportunity type. We believe, however, that our methodology can be extended to other opportunity types essential for a meaningful life, such as healthcare and education, provided that the analyst has the necessary data on hand to perform the analysis.

Although the NSTP provides an unprecedented snapshot of transport poverty across Canada, the survey does not contain a question in which the respondent answers objectively whether they are satisfied or not with their mobility experiences or overall transport conditions. Rather, More specifically, the respondents use a continuum scale to answer their level of agreement with specific statements. The absence of a categorical or binary variable referring to satisfaction level led us to categorize the ~~outcomes~~outcome, in order to identify which ~~perception-response~~ category the respondent belongs to, varying from most negative to most positive. In fact, with the information we have, we cannot infer if the person is satisfied or not, but rather whether they are more or less satisfied in comparison to others. The underlying rationale for our standards ~~proposition is to identify the calculated accessibility level that places~~ is therefore relative: they identify the minimum objective accessibility at which the most disadvantaged ~~population group within the most positive perception-group~~ profile is no longer more likely than not to be among the respondents reporting the most negative perceptions in the region.

It is worth mentioning that other outcome categorizations or numbers of categories are possible. For instance, one might decide to convert the continuum scale to a Likert scale based on fixed intervals, classifying those with agreement levels ~~between from~~ 0 to 20% as “Most Negative”, 20 to 40% as “Negative”, and so on, in pre-determined intervals. The problem with ~~this the fixed-interval~~ approach is that ~~when doing this~~, the analyst is normatively defining levels of satisfaction for ~~the~~ respondents that might not ~~represent the actual feeling of the person~~ reflect their actual feelings. Because of this, we decided to categorize responses based on the perception scale distribution and use the sample population to evaluate those who are more and less satisfied.

~~This outcome~~ To test the sensitivity of our results to the choice of categorization type, we estimated a new perception model using a fixed-interval categorization that does not depend on the sample’s distribution, instead of a quintile-based categorization (classifying responses from 0 to 20 as most positive, 20 to 40 as positive, 40 to 60 as neutral, 60 to 80 as negative, and 80 to 100 as most negative). The results of this alternative model (Appendix A.4) are consistent with those of the model created using the quintile classification. The accessibility estimates retain their signs and similar magnitudes (e.g., transit accessibility: $OR = 1.30$, $p = 0.027$, compared to $OR = 1.25$, $p = 0.044$ in our main specification; and active accessibility: $OR = 0.87$, $p = 0.041$, compared to $OR = 0.86$, $p = 0.024$), as do the statistically significant control variables. The only coefficient that loses statistical significance is the interaction between active accessibility and being an active traveller ($OR = 1.13$, $p = 0.10$, compared to $OR = 1.17$, $p = 0.029$). Its point estimate remains similar, which suggests a loss of precision rather than a change in direction.

However, the quintile-based categorization caused, in general, the control variables to lose part of their predictive power when compared to linear regression models created using the previous continuous forms. As shown in Table A.3, in general, ~~the accessibility variables presented a similar behaviour when compared and similarly~~ to the categorical models; ~~However, the accessibility estimates are consistent with our hypothesized causal effect. In contrast~~, the control variables presented a better performance in explaining the outcomes in the linear regressions, such as in the case of the disability indicator and the “temporary visa or refugee” variable, which did not present statistical significance in the categorical version of ~~the model for “there are too few jobs available for me in my ideal commuting time.”~~ We decided to use the categorical models because they enabled us to identify levels of satisfaction in a non-normative way perception model.

An additional limitation concerns the interpretation of our declined job offers indicator. Agreeing or disagreeing with this question presupposes that the respondent had looked for a job and subsequently received a job offer that they declined. In this sense, individuals who did not look for work (including those who may have given up searching because they anticipated that available jobs would be inaccessible due to their transportation situation) would not be recorded as having declined an offer, regardless of the extent to which transportation constraints limit their employment prospects. This mechanism is conceptually related to the discussion of adaptive preferences: just as some people may adjust downward their perceptions of job scarcity in response to low accessibility, others may give up on the job search completely. Consequently, the indicator of declined job offers likely captures transportation-related employment disadvantage only among those who remain active in the labour market and may underestimate the prevalence of transportation disadvantage among those most severely excluded from employment opportunities. We cannot assess this selection directly using the current cross-sectional data from the NSTP, which do not allow us to distinguish those who are not looking for work from those who are actively seeking employment but have not received offers to decline.

Regarding the relationship between perceived and ~~calculated accessibility, only transit accessibility, in one of the models, presented the behaviour hypothesized in~~ objective accessibility, the transit accessibility indicator was the most consistent with our hypothesized causal effect in the perception-based model, through which increasing accessibility was associated with more positive perceptions. Although the active accessibility indicator presented a slightly positive net ~~effect association~~ for active travellers in both models, this effect was not sufficient to outweigh the influence of the other explanatory variables. As a result, the estimated active thresholds were excessively demanding and unrealistic from a policy perspective. For one model, we were unable to identify thresholds, while for the other, the conservative threshold was 22 times higher than the median accessibility level, and the progressive threshold was more than 123 times higher than the median accessibility level. These results led us to propose equity standards ~~based solely on the model “there~~

~~are too few jobs available for me in my ideal commuting time,”~~, exclusively, based on the perception-based model and for public transit mode.

Additionally, the non-significant residuals in the categorical models indicate weak evidence of endogeneity in the objective accessibility indicators. However, a Wu-Hausman test on the linear (continuous) version of the employment-outcome model did detect significant endogeneity ($p < 0.05$; Appendix A.2), suggesting that the categorical models’ failure to detect it may instead reflect a loss of statistical power from quintile-categorization of the dependent variables. This result reinforces our view that the accessibility variables are unlikely to be entirely exogenous to the perceived accessibility indicators and the outcomes analyzed, and that the instrumental variable framework remains necessary. As is generally the case in instrumental variable designs (Cunningham, 2021; Huntington-Klein, 2021), our assumption that the instruments affect perceived accessibility and employment outcome only through objective accessibility cannot be directly tested. Even so, adopting an instrumental variable approach was the most suitable alternative for accounting for potential endogeneity given the use of cross-sectional surveys.

Our instrumental variable strategy was not specifically designed to address residential self-selection, and although it offers a degree of protection against confounding factors more broadly, we cannot exclude the possibility that unobserved commuting preferences and constraints that determine residential location choices are correlated with our spatial filter instruments. A more definitive test would require longitudinal data tracking individuals across their residential moves over the years, or an instrument based on a source of variation known to be exogenous to residential choice; neither of these options is available in the current cross-sectional NSTP data, nor in the Canadian Census.

For future research, we suggest exploring additional place-based-objective accessibility measures to further evaluate the relationship between ~~calculated-objective~~ and perceived accessibility. For instance, the raw multimodal spatial availability indicator could be tested. This indicator ~~uses as output reflects~~ the number of jobs accessible by ~~different transport modes, not standardized each transport mode, without standardizing~~ by the number of people ~~using these modes competing for those jobs~~. We decided to use multimodal spatial availability per person because we believe that competition matters when evaluating perceptions related to employment outcomes, as demonstrated by previous studies (Jin and Paulsen, 2018; Merlin and Hu, 2017; Miller, 2005). ~~Another limiting factor that led us to adopt the CT level was that we could not estimate travel times at the DA level for the entire study area due to the memory limitations of our computational resources. Despite that, the initial tests performed with a smaller study area did not show differences in terms of the sign and statistical significance of the accessibility coefficients between regressions – probably because the indicators are standardized; and because the standardized version allows for comparisons across cities with different labour force and job opportunity sizes.~~

~~Continuing with~~ Regarding the spatial unit used, the CTs, we recognize that this unit is not the most detailed among census geographies, being less detailed than dissemination areas or dissemination blocks. We decided to use CTs due to the large size of the study area, in order to provide a regional analysis rather than focus on only one city, considering this entire basin to calculate the accessibility indicators and select the sample. This choice is motivated by the recognition of commuting flows across municipalities, ~~which are even a pattern that is itself~~ one of the parameters used by Statistics Canada when defining Census Metropolitan Areas.

~~Regarding the equity standards definitions, three moral principles guided our approach. The first relies on the recognition of accessibility as a human capability (Sen, 2005, 1999; Pereira et al., 2017; Luz and Portugal, 2022) – something required for a dignified life (Karner et al., 2020), and the main transportation benefit to be distributed across society in order to increase people’s freedom. The second is grounded in the sufficientarian perspective, considering a fair transportation and land-use system as one that provides sufficient accessibility to all members of society (Martens and Golub, 2021) as requirement for them to meet their basic needs (Pereira et al., 2017; Pereira and Karner, 2021). The third moral principle is grounded in an egalitarian perspective, in which the minimum level of accessibility is defined in terms of the most equity-deserving population groups to reduce inequalities of opportunities. Making explicit the moral foundations that support our analysis is, in our perspective, important because different theories of justice lead to different perspectives on how policies should manage the distribution of transport-related benefits.~~

~~Transport equity is only one step towards achieving transport justice (Karner et al., 2020): a condition~~

in which no person or group is disadvantaged by a lack of access to the opportunities they need to lead a dignified and healthy life. To achieve this state, it is necessary to transform the structures and processes that lead to the unequal distribution of the multiple externalities of transport across population groups and space. In this study, we propose a methodology to suggest equity standards based on a data-driven approach. We intend to suggest minimum levels of accessibility that could guide transportation planning towards a more equitable distribution of transportation benefits, as an earlier stage to be succeeded by a participatory and democratic process to collectively select the best standard, considering that residents and other stakeholders must be able to actively participate in and influence the decisions that affect their lives. While this is, in our perspective, the most appropriate method for standards definition, it is beyond the scope of this study to develop democratic plenaries to discuss the suggested standards.

A positive aspects from this research is that the equity standards are suggested based on their link with perceived accessibility, making them easier to communicate with policy practitioners and the general public. We also accounted for possible endogeneity of calculated accessibility, so that the coefficients represent the true effect of the indicators and not a mere correlation. Although the study does not ensure full reproducibility, since the used surveys are confidential, it ensures replicability by making all codes available for consultation. Another limiting factor that led us to adopt the CT level was that we could not estimate travel times at the dissemination area level for the entire study area due to the memory limitations of our computational resources. Despite that, the initial tests performed with a smaller study area (for the City of Toronto) did not show differences in terms of the sign and statistical significance of the accessibility coefficients between regressions - probably because the indicators are standardized based on the labour force size.

6. Conclusions

The main objective of this study was to ~~suggest~~ propose a model-based methodology to define equity standards for transportation planning, based on the relationship between ~~calculated job accessibility and perceived accessibility~~ referred to employment outcomes, for a study area covering objective job accessibility, people's perceived accessibility, and an employment outcome. The study area covers municipalities around the cities of Toronto and Hamilton, in southern Ontario, Canada. ~~For that~~ To this end, we used data from the 2021 Census of Population and the recently released National Survey on Transport Poverty (Tiznado-Aitken et al., 2025), combining (place-based ~~calculated~~) objective accessibility measures with variables at the individual level, ~~while controlling for endogeneity using instrumental variables in a two-stage regression estimation.~~

~~The place-based~~ Our approach is distinguished from other studies by translating estimated associations between objective accessibility, perceived accessibility, and an employment outcome into planning thresholds and by proposing thresholds based on objective accessibility calculated from spatial land-use and transport data and validating them against how accessibility is actually experienced by residents. Although the proposed standards are presented at the neighbourhood level, they indirectly also incorporate individual-level characteristics, since these standards are obtained from individuals' self-reported perceptions and outcomes in the NSTP. In this way, these thresholds can assist policymakers in interpreting how different levels of objective accessibility are associated with residents' perceived ability to reach employment opportunities under existing social and spatial constraints. This increases the potential for government adoption and enhances the comprehensibility and relevance of such standards for community members and those experiencing unequal accessibility. Furthermore, identifying sufficient job accessibility levels can help assess and highlight areas at higher risk of transport poverty, particularly those with a higher presence of transport-disadvantaged populations.

Objective accessibility models were created using multimodal spatial availability per person, considering walking and cycling (later combined as active transportation), public transit, and ~~motorized transportation~~ modes. In general, our models showed that these indicators partially predicted with statistical significance perceptions related to how easy it is to reach work and declining employment opportunities because of transport situation. For the model regarding the perception of having too few jobs available within an ideal commuting time, ~~private vehicles.~~

Both models were statistically significant overall (Wald F-tests, $p < 0.001$). In the perceived accessibility model, the public transit accessibility behaved according to our hypothesis that increasing calculated accessibility would reduce negative perceptions, in which one additional job accessible by public transit estimates are consistent with our hypothesized relationship, with a one-standard-deviation increase in transit accessibility (about 0.09 jobs per transit user) increases the average associated with 25% higher odds of being in a more positive perception group by 25% (p-value response category ($p < 0.05$)). Our models also showed (p-value < 0.05) that, for active users, active accessibility has For active travellers, the estimates suggest a slightly positive effect on individual perception, increasing their odds of being in a positive category. Using the “few jobs available” association between active accessibility and both outcomes, although this combined association is small.

Using the perception-based model and for the public transit mode, we suggest equity standards based on the accessibility threshold that results in a higher likelihood of positive perceptions for people from equity-deserving transport-disadvantaged groups. We suggest three equity standards: a conservative standard (0.16 jobs reached by public transit per transit user), an intermediate standard (0.26), and a progressive standard (0.36).

Our results show that most of the study area falls short of the proposed equity standards, regardless of their category (conservative, intermediate, or progressive). About 66% of the study area population lives and 55% of the low-income population live in areas with transit accessibility below the conservative standard, while 91% of the general population and 87% of the low-income population live in areas below the intermediate standard, and almost the entire study area population lives below the progressive standard.

Finally, we believe our study contributes to the ongoing discussion of equity in transportation, and that equity standards based on, or inspired by, our methodology can serve as a data-driven model-based starting point for debate among residents and other stakeholders within a participatory and democratic process to collectively define the most appropriate standard.

CRediT authorship contribution statement

(include after the review)

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Data availabilityOpen science statement

We created To facilitate collaboration and further analysis and to ensure the transparency and reproducibility of our study, we developed this paper using literate programming in which the R markdown code to fully reproduce this article is available on our GitHub repository (<https://github.com/dias-bruno/NSTP/>), with the data analysis code accessible through our GitHub page (<https://github.com/dias-bruno/NSTP/>), in alignment with best practices in open spatial data science (Páez, 2021; Arribas-Bel et al., 2021).

Declaration of competing interest

The authors declare no conflicts of interest.

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Appendix

Table A.1: Impedance functions calibrated for the census ~~subdivisions~~ subdivisions of the study area.

Census Division	Census Code	Mode	Impedance function	Parameter 1*	Parameter 2*	AIC
Durham	3518	Car-motor	Gamma	1.92	0.07	1714717
	3518	Walk	Exponential	0.09		42507
	3518	Transit	Gamma	3.33	0.06	121000
	3518	Bike	Gamma	1.89	0.08	3829
York	3519	Car-motor	Gamma	2.16	0.08	2549528
	3519	Transit	Normal	54.89	28.44	200698
	3519	Walk	Exponential	0.09		58149
	3519	Bike	Lognormal	2.75	0.81	8060
Toronto	3520	Transit	Gamma	3.60	0.08	1880937
	3520	Car-motor	Gamma	2.33	0.09	3942809
	3520	Walk	Gamma	1.33	0.08	449370
	3520	Bike	Lognormal	2.94	0.65	123658
Peel	3521	Car-motor	Gamma	2.33	0.09	3351993
	3521	Walk	Exponential	0.08		72360
	3521	Bike	Lognormal	2.87	0.73	8853
	3521	Transit	Gamma	3.28	0.06	468686
Halton	3524	Car-motor	Gamma	1.92	0.07	1264978
	3524	Transit	Normal	53.20	26.65	75851
	3524	Walk	Lognormal	1.90	1.08	40360
	3524	Bike	Gamma	1.99	0.11	6229
Hamilton	3525	Car-motor	Gamma	1.85	0.07	1323879
	3525	Walk	Exponential	0.08		61816
	3525	Transit	Gamma	2.24	0.05	128266
	3525	Bike	Lognormal	2.83	0.75	9830

Note:

For each probability distribution, Parameter 1 and Parameter 2 correspond to their standard defining parameters: the mean and standard deviation (for lognormal and normal), the rate and shape (for gamma), and the power (for exponential). The AIC represents the Akaike Information Criterion used to assess model fit.

Table A.2: Instrument relevance (weak-instrument F-tests) and endogeneity (Wu–Hausman) diagnostics for linear (2SLS) models estimated on the original continuous dependent variables.

Diagnostic test	Few jobs		Declined job offers	
	F-statistic	p-value	F-statistic	p-value
Weak instruments: active accessibility	66.15***	7.07e-64	127.93***	4.89e-109
Weak instruments: transit accessibility	358.63***	1.02e-271	232.82***	1.26e-174
Weak instruments: private vehicle accessibility	498.25***	0	324.19***	5.72e-221
Weak instruments: active $\times$ user	18.60***	4.33e-18	13.41***	9.39e-13
Weak instruments: transit $\times$ user	35.06***	2.06e-34	13.64***	5.63e-13
Wu–Hausman (endogeneity test)	1.29	0.263	2.59*	0.0245

*p<0.05; **p<0.01; ***p<0.001

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Table A.3: Weighted linear regression models connecting ~~calculated-objective~~ to perceived accessibility and to employment outcome.

Variable	Few jobs			Declined job offers		
	Beta	SE	p-value	Beta	SE	p-value
(Intercept)	79***	7.42	<0.001	50***	9.31	<0.001
Predicted active accessibility per user	2.1	3.10	0.5	13**	4.61	0.006
Predicted transit accessibility per user	-76***	18.2	<0.001	-5.1	26.3	0.8
Predicted private vehicle accessibility per user	-6.6	5.20	0.2	-9.2	7.05	0.2
Number of Cars	0.65	1.54	0.7	0.20	1.62	>0.9
Number of Bikes	-1.5	0.947	0.11	-0.16	1.06	0.9
Preferred mode match most frequent mode	-0.29	2.10	0.9	2.1	2.62	0.4
Monthly or annual public transit pass	5.1*	2.35	0.030	21***	3.62	<0.001
Public transit availability in the neighbourhood	6.8*	3.39	0.045	-5.2	3.22	0.11
Indigenous or racialized person	12***	2.20	<0.001	14***	2.71	<0.001
Temporary visa or refugee	-9.1*	3.86	0.019	8.8	5.52	0.11
Person with disability	6.7**	2.39	0.005	-3.2	2.97	0.3
Woman, transgender or non-binary person	2.3	1.96	0.2	-0.47	2.45	0.8
Age	-0.16*	0.081	0.041	-0.21*	0.089	0.017
Household size	-1.3	0.653	0.054	1.4	0.884	0.12
Household income	***		<0.001	***		<0.001
Less than \$40,000	—	—		—	—	
\$40,000 to \$80,000	1.7	2.34		-1.4	2.95	
\$80,000 to \$120,000	-7.5	2.85		-9.2	3.57	
More than \$120,000	-9.9	2.79		-22	3.56	
Population density (people/km2)	0.00**	0.000	0.006	0.00	0.000	0.4
Population centre class	**		0.008	***		<0.001
Rural	—	—		—	—	
Small	-8.5	7.41		11	7.53	
Medium	-18	5.95		-19	5.41	
Large	-1.3	3.86		1.2	4.78	
Active mode interaction: accessibility × user	-12*	5.47	0.027	-4.1	5.48	0.5
Transit mode interaction: accessibility × user	22	15.5	0.2	23	24.7	0.3

Abbreviation: SE = Standard Error

*p<0.05; **p<0.01; ***p<0.001

Table A.4: [Alternative weighted ordinal logistic regression model connecting objective to perceived accessibility using a fixed-interval categorization for the dependent variable.](#)

Variable	OR	SE	p-value
Active job accessibility per active traveller	0.87*	0.069	0.041
Active mode interaction: accessibility $\times$ user	1.13	0.076	0.10
Transit job accessibility per transit user	1.30*	0.118	0.027
Transit mode interaction: accessibility $\times$ user	0.96	0.138	0.8
Private vehicle accessibility per user	0.92	0.099	0.4
Number of Cars	0.97	0.109	0.8
Number of Bikes	1.07	0.063	0.3
Preferred mode match most frequent mode	1.04	0.176	0.8
Monthly or annual public transit pass	0.68*	0.169	0.022
Public transit availability in the neighbourhood	0.75	0.225	0.2
Indigenous or racialized person	0.61**	0.172	0.004
Temporary visa or refugee	1.51	0.255	0.11
Person with disability	0.73	0.197	0.11
Woman, transgender or non-binary person	0.93	0.150	0.7
Age	1.01	0.006	0.057
Household size	1.10	0.054	0.066
Household income			0.2
Less than \$40,000	—	—	
\$40,000 to \$80,000	0.86	0.181	
\$80,000 to \$120,000	1.28	0.219	
More than \$120,000	1.30	0.239	
Population density (people/km ²)	1.19*	0.076	0.023
Population centre class			0.5
Rural	—	—	
Small	1.14	0.630	
Medium	2.07	0.466	
Large	1.16	0.307	
Residuals: Active accessibility	1.09	0.083	0.3
Residuals: Transit accessibility	0.85	0.108	0.14
Residuals: Private vehicle accessibility	1.13	0.067	0.078
Wald F	2.90		
p-value	<0.001		
AIC	6,571		

Abbreviations: OR = Odds Ratio, SE = Standard Error

*p<0.05; **p<0.01; ***p<0.001. Accessibility indicators, population density, and first-stage residuals are standardized; odds ratios correspond to a one-standard-deviation increase.

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